✦ DATA PROFILE // ELESSAR

UCI Wine Chemistry.

178 rows × 14 columns · pandas.DataFrame · read in 0.0s · generated 2026-11-02 07:59 by elessar 1.1.0

ROWS	COLUMNS	MISSING CELLS	ALERTS	TARGET
178	**14**	**0.0%**	**5**	**cultivar**
18.7 KB in memory	2 semantic types	0 rows with one	0 high · 1 medium · 4 info	the column the others explain

01 / KEY FINDINGS

Key findings.

- `flavanoids` is strongly associated with `cultivar` (eta 0.853, p_adj 6e-28)
- `proline` is strongly associated with `cultivar` (eta 0.839, p_adj 9e-26)
- `od280/od315_of_diluted_wines` is strongly associated with `cultivar` (eta 0.827, p_adj 1e-23)
- MEDIUM rows are ordered (sorted file or time series): lag-1 autocorrelation flavanoids +0.79, od280/od315_of_diluted_wines +0.76, color_intensity +0.74, proline +0.74, hue +0.69, alcohol +0.65 (+6 more) → shuffle before a random split, or use time-aware validation

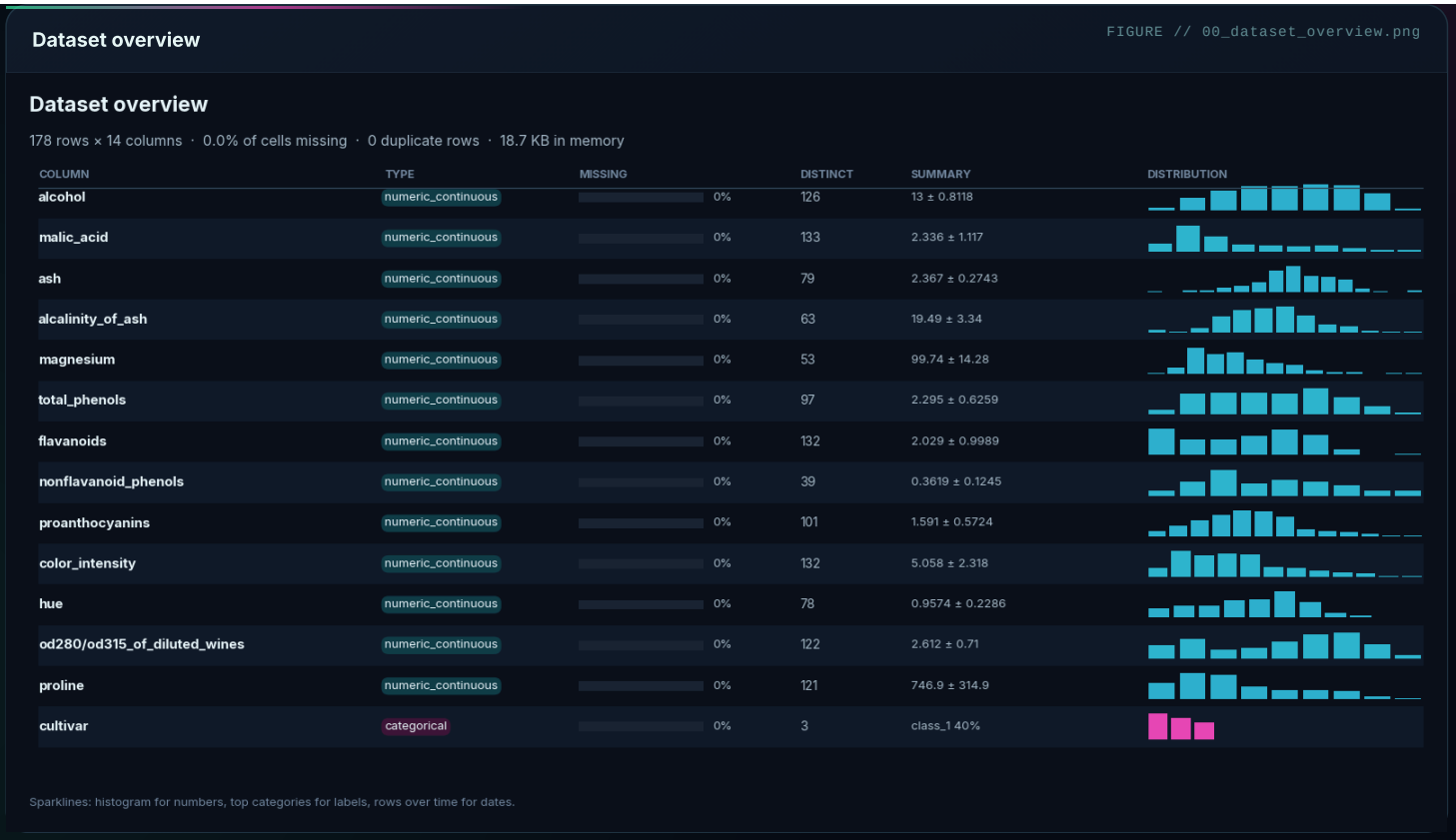
02 / OVERALL STATISTICS

Overall statistics.

Rows	178
Columns	14
Memory	18.7 KB
Potential saving	9.0 KB
Pandas dtypes	float64=13, category=1
Semantic types	numeric_continuous=13, categorical=1
Duplicate rows	0 (0.0%)
Missing cells	0.0%
Target	cultivar

Dataset overview

FIGURE // 00_dataset_overview.png



Numeric columns at a glance

COLUMN	COUNT	MISSING	MEAN	STD	MIN	MEDIAN	MAX	SKEW	KURTOSIS	OUTLIERS	BEST FIT
alcohol	178	0.0%	13	0.8118	11.03	13.05	14.83	-0.0515	-0.852	0.0%	norm(loc=13, scale=0.81)
malic_acid	178	0.0%	2.336	1.117	0.74	1.865	5.8	1.04	0.299	1.7%	lognorm(s=0.449, scale=2.11)
ash	178	0.0%	2.367	0.2743	1.36	2.36	3.23	-0.177	1.14	1.7%	logistic(loc=2.37, scale=0.151)
alcalinity_of_ash	178	0.0%	19.49	3.34	10.6	19.5	30	0.213	0.488	2.2%	logistic(loc=19.4, scale=1.87)
magnesium	178	0.0%	99.74	14.28	70	98	162	1.1	2.1	2.2%	lognorm(s=0.136, scale=98.8)
total_phenols	178	0.0%	2.295	0.6259	0.98	2.355	3.88	0.0866	-0.836	0.0%	weibull_min(c=4.12, scale=2.53)
flavanoids	178	0.0%	2.029	0.9989	0.34	2.135	5.08	0.0253	-0.88	0.0%	beta(a=1.25, b=2.27, loc=0.335, scale=4.75)
nonflavanoid_phenols	178	0.0%	0.3619	0.1245	0.13	0.34	0.66	0.45	-0.637	0.0%	gamma(a=8.42, scale=0.043)
proanthocyanins	178	0.0%	1.591	0.5724	0.41	1.555	3.58	0.517	0.555	1.1%	gamma(a=7.26, scale=0.219)

COLUMN	COUNT	MISSING	MEAN	STD	MIN	MEDIAN	MAX	SKEW	KURTOSIS	OUTLIERS	BEST FIT
color_intensity	178	0.0%	5.058	2.318	1.28	4.69	13	0.869	0.382	2.2%	lognorm(s=0.459, scale=4.56)
hue	178	0.0%	0.9574	0.2286	0.48	0.965	1.71	0.0211	-0.344	0.6%	norm(loc=0.957, scale=0.228)
od280/od315_of_diluted_wines	178	0.0%	2.612	0.71	1.27	2.78	4	-0.307	-1.09	0.0%	beta(a=1.21, b=1.33, loc=1.27, scale=2.74)
proline	178	0.0%	746.9	314.9	278	673.5	1,680	0.768	-0.248	0.0%	lognorm(s=0.414, scale=686)

Categorical columns at a glance

COLUMN	TYPE	COUNT	MISSING	DISTINCT	MOST FREQUENT	SHARE	ENTROPY
cultivar	categorical	178	0.0%	3	class_1	39.9%	0.989

How each column was typed

Each column goes down a decision tree of explicit questions. Certain: every threshold test is decided at 95% confidence (Wilson interval of the observed share) and no label is likely unseen (Good-Turing unseen mass below 1%). Probable: an identifier, empty or constant column, which only every row can prove. Uncertain: a test too close to its threshold, too few values, or an ambiguous date order.

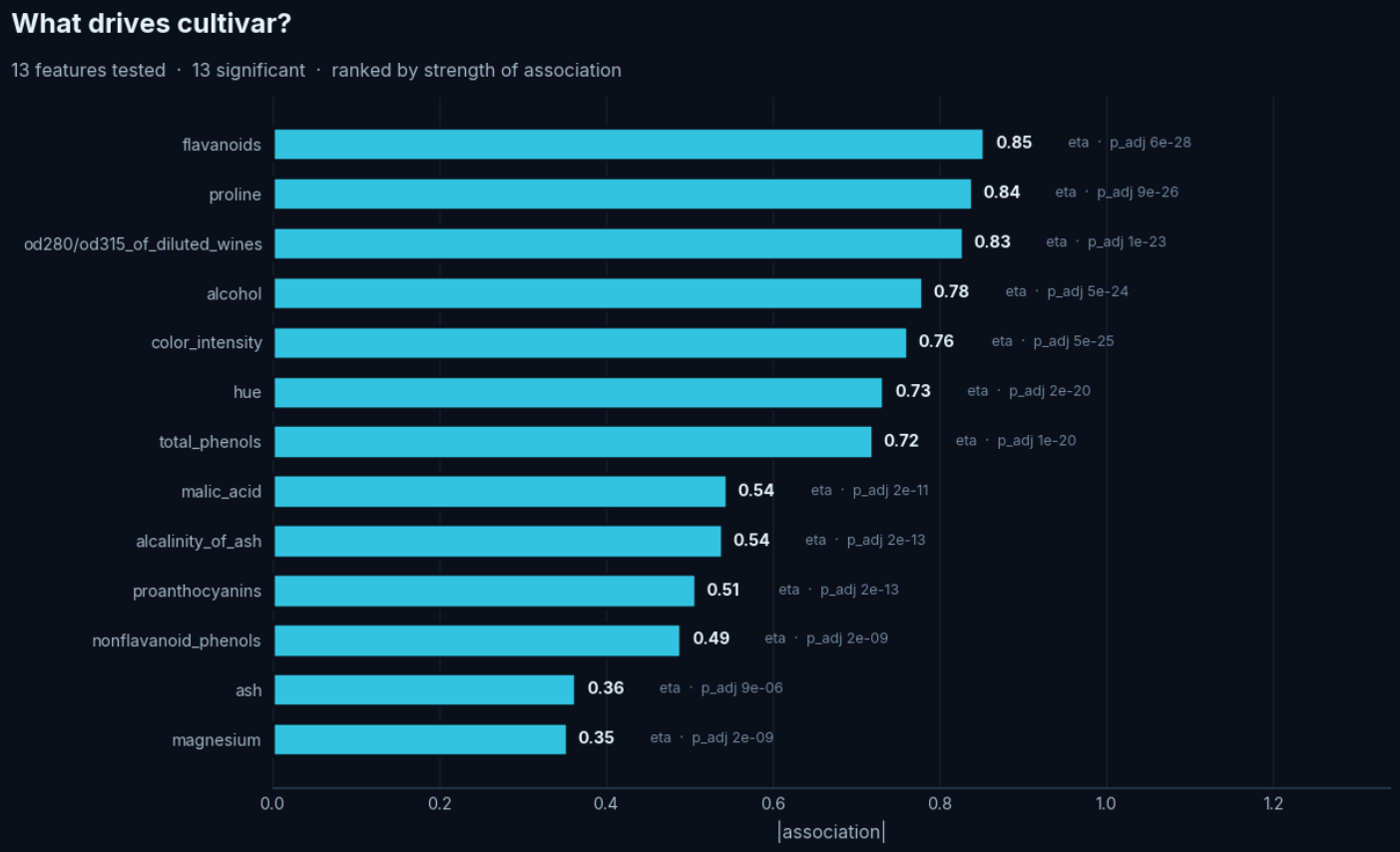
COLUMN	TYPE	CERTAINTY	WHY	DECIDED BY	QUESTIONS
alcohol	numeric_continuous	CERTAIN	decided on every value	real numbers	12
malic_acid	numeric_continuous	CERTAIN	decided on every value	real numbers	12
ash	numeric_continuous	CERTAIN	decided on every value	real numbers	12
alcalinity_of_ash	numeric_continuous	CERTAIN	decided on every value	real numbers	12
magnesium	numeric_continuous	CERTAIN	decided on every value	integers?	10
total_phenols	numeric_continuous	CERTAIN	decided on every value	real numbers	12
flavanoids	numeric_continuous	CERTAIN	decided on every value	real numbers	12
nonflavanoid_phenols	numeric_continuous	CERTAIN	decided on every value	real numbers	12
proanthocyanins	numeric_continuous	CERTAIN	decided on every value	real numbers	12
color_intensity	numeric_continuous	CERTAIN	decided on every value	real numbers	12
hue	numeric_continuous	CERTAIN	decided on every value	real numbers	12
od280/od315_of_diluted_wines	numeric_continuous	CERTAIN	decided on every value	real numbers	12
proline	numeric_continuous	CERTAIN	decided on every value	integers?	10
cultivar	categorical	CERTAIN	decided on every value	how is it stored? (category)	3

Target analysis.

Target: `cultivar` (categorical)

cultivar · what drives the target

FIGURE // 21_cultivar_target_ranking.png



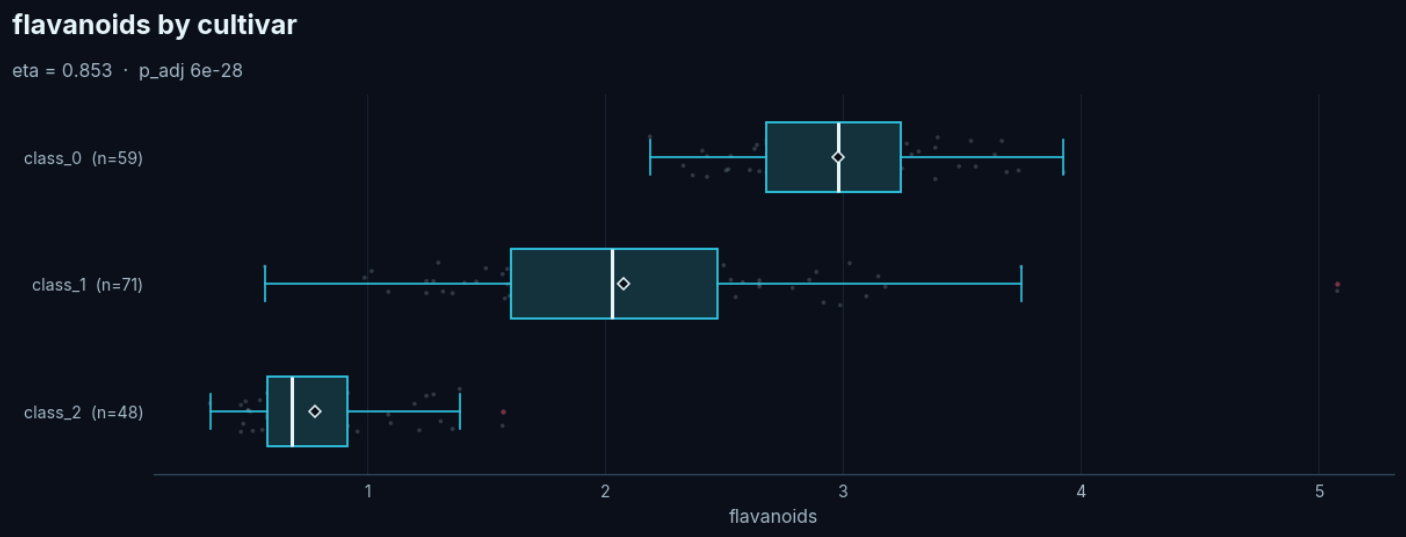
Association: Spearman (numbers), eta (numbers vs labels), Cramér's V (labels). Grey = not significant after Benjamini-Hochberg correction (α = 0.05).

FEATURE	TYPE	METHOD	ASSOCIATION	TEST	N	P_ADJ	SIG	AUC	MI	FLAG
flavanoids	numeric_continuous	eta	+0.853	kruskal-wallis	178	6e-28	*		0.971	
proline	numeric_continuous	eta	+0.839	kruskal-wallis	178	9e-26	*		0.836	
od280/od315_of_diluted_wines	numeric_continuous	eta	+0.827	kruskal-wallis	178	1e-23	*		0.769	
alcohol	numeric_continuous	eta	+0.779	kruskal-wallis	178	5e-24	*		0.676	
color_intensity	numeric_continuous	eta	+0.761	kruskal-wallis	178	5e-25	*		0.810	
hue	numeric_continuous	eta	+0.733	kruskal-wallis	178	2e-20	*		0.652	
total_phenols	numeric_continuous	eta	+0.719	kruskal-wallis	178	1e-20	*		0.587	
malic_acid	numeric_continuous	eta	+0.545	kruskal-wallis	178	2e-11	*		0.448	
alcalinity_of_ash	numeric_continuous	eta	+0.539	kruskal-wallis	178	2e-13	*		0.369	
proanthocyanins	numeric_continuous	eta	+0.507	kruskal-wallis	178	2e-13	*		0.352	

FEATURE	TYPE	METHOD	ASSOCIATION	TEST	N	P_ADJ	SIG	AUC	MI	FLAG
nonflavanoid_phenols	numeric_continuous	eta	+0.490	kruskal-wallis	178	2e-09	*		0.287	
ash	numeric_continuous	eta	+0.363	kruskal-wallis	178	9e-06	*		0.235	
magnesium	numeric_continuous	eta	+0.353	kruskal-wallis	178	2e-09	*		0.333	

cultivar vs flavanoids · target by feature

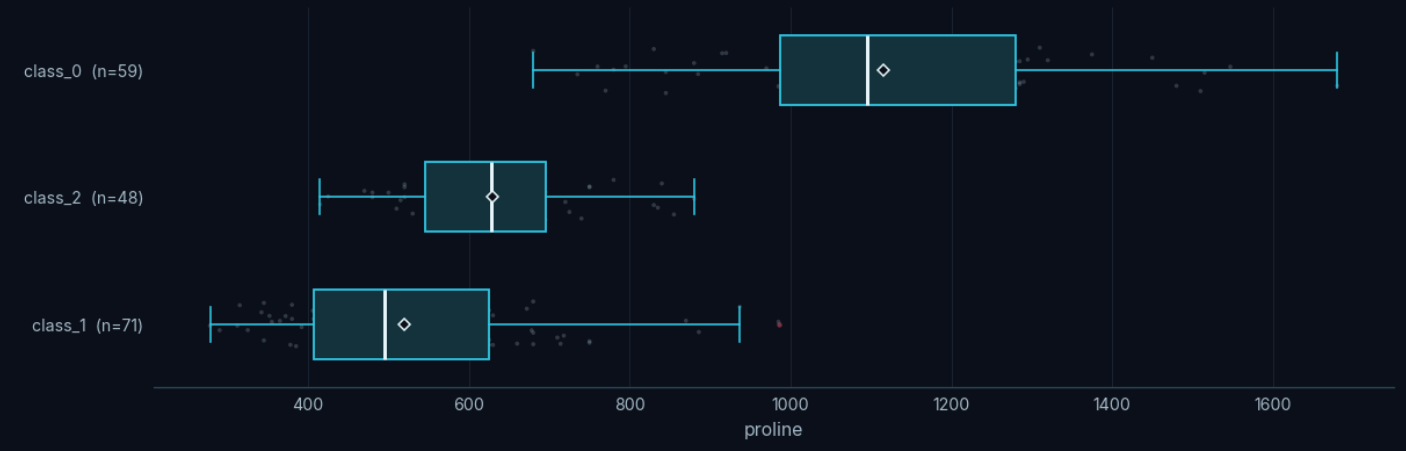
FIGURE // 22_cultivar__flavanoids_target.png



Box = interquartile range, line = median, diamond = mean, dots = individual rows (sampled); groups sorted by median.

cultivar vs proline · target by feature

FIGURE // 23_cultivar__proline_target.png



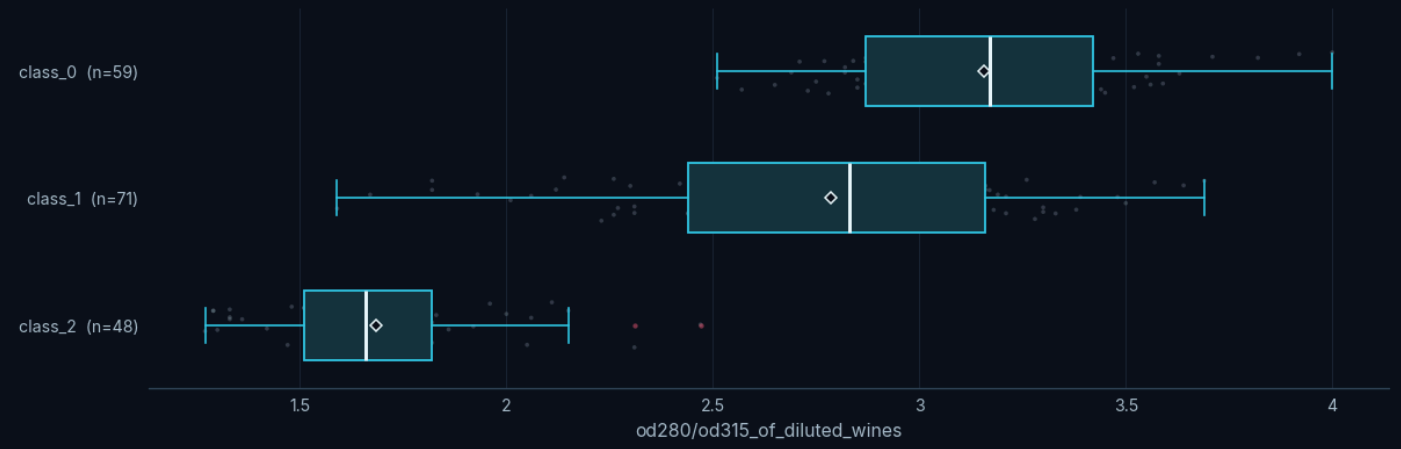
Box = interquartile range, line = median, diamond = mean, dots = individual rows (sampled); groups sorted by median.

cultivar vs od280/od315_of_diluted_wines · target by feature

FIGURE // 24_cultivar__od280_od315_of_diluted_wines_target.png

od280/od315_of_diluted_wines by cultivar

eta = 0.827 · p_adj 1e-23



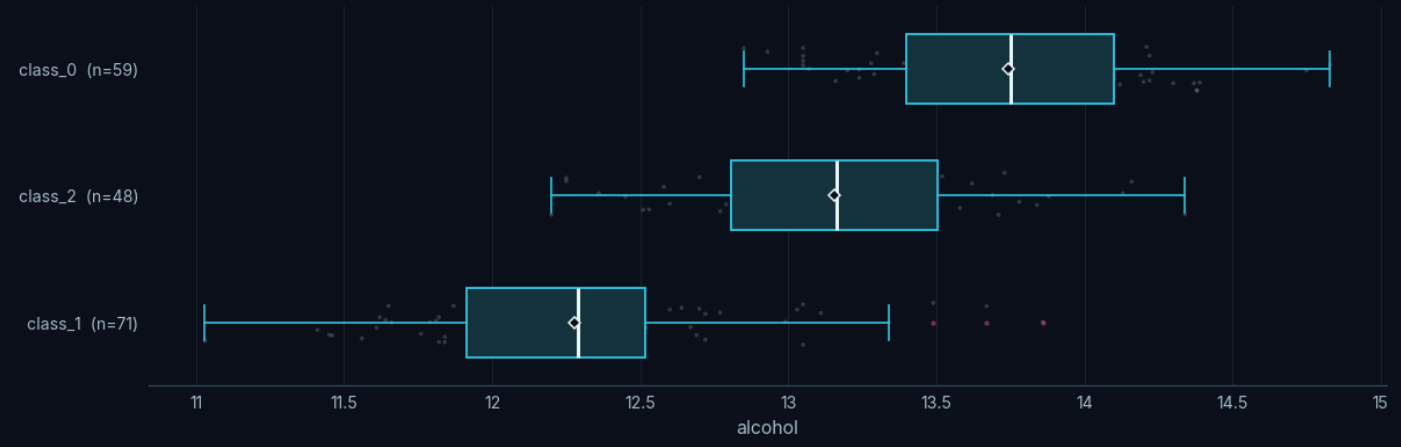
Box = interquartile range, line = median, diamond = mean, dots = individual rows (sampled); groups sorted by median.

cultivar vs alcohol · target by feature

FIGURE // 25_cultivar__alcohol_target.png

alcohol by cultivar

eta = 0.779 · p_adj 5e-24



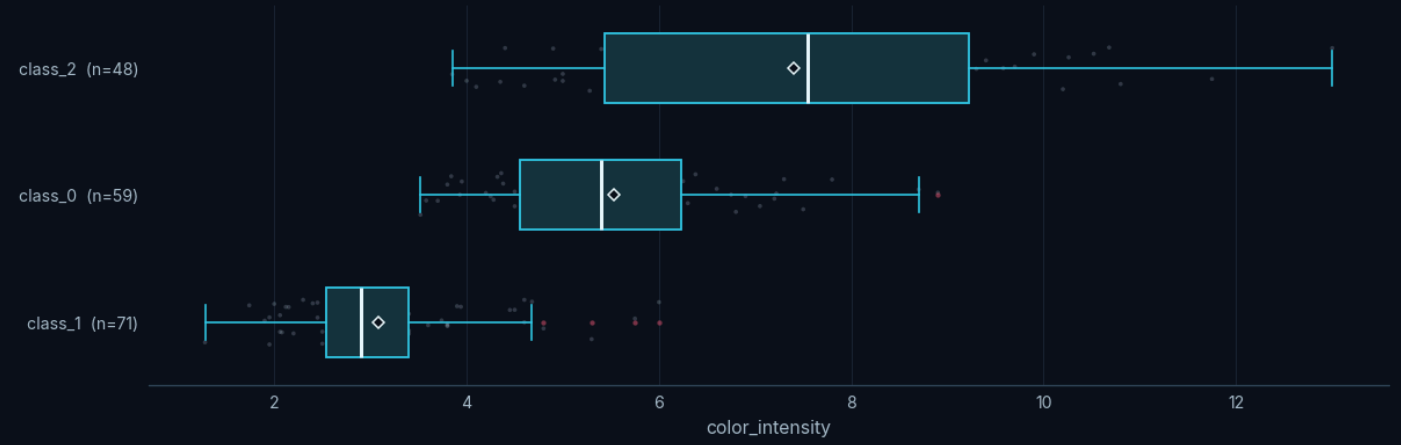
Box = interquartile range, line = median, diamond = mean, dots = individual rows (sampled); groups sorted by median.

cultivar vs color_intensity · target by feature

FIGURE // 26_cultivar__color_intensity_target.png

color_intensity by cultivar

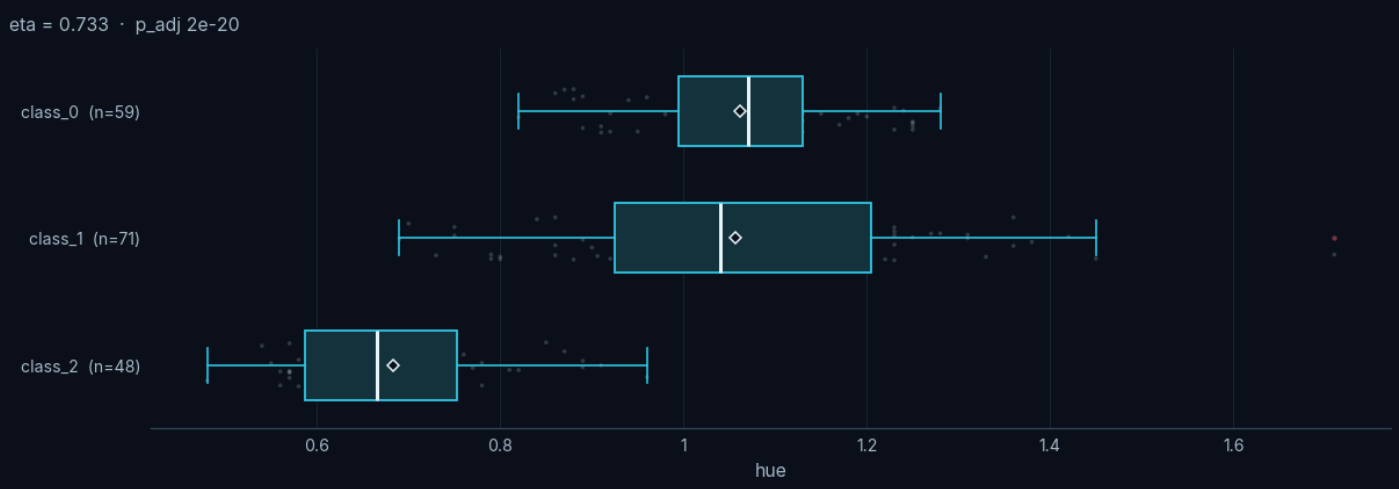
eta = 0.761 · p_adj 5e-25



Box = interquartile range, line = median, diamond = mean, dots = individual rows (sampled); groups sorted by median.

cultivar vs hue · target by feature

FIGURE // 27_cultivar__hue_target.png



Box = interquartile range, line = median, diamond = mean, dots = individual rows (sampled); groups sorted by median.

04 / FEATURE SELECTION

Feature selection.

3-class classification of cultivar on 178 rows and 13 features (13 model columns after one-hot encoding). Held-out balanced_accuracy: decision tree 0.944, random forest 1.000, lasso 0.967, ridge 1.000 (tuned by 5-fold CV on 142 rows, scored on 36 held-out rows).

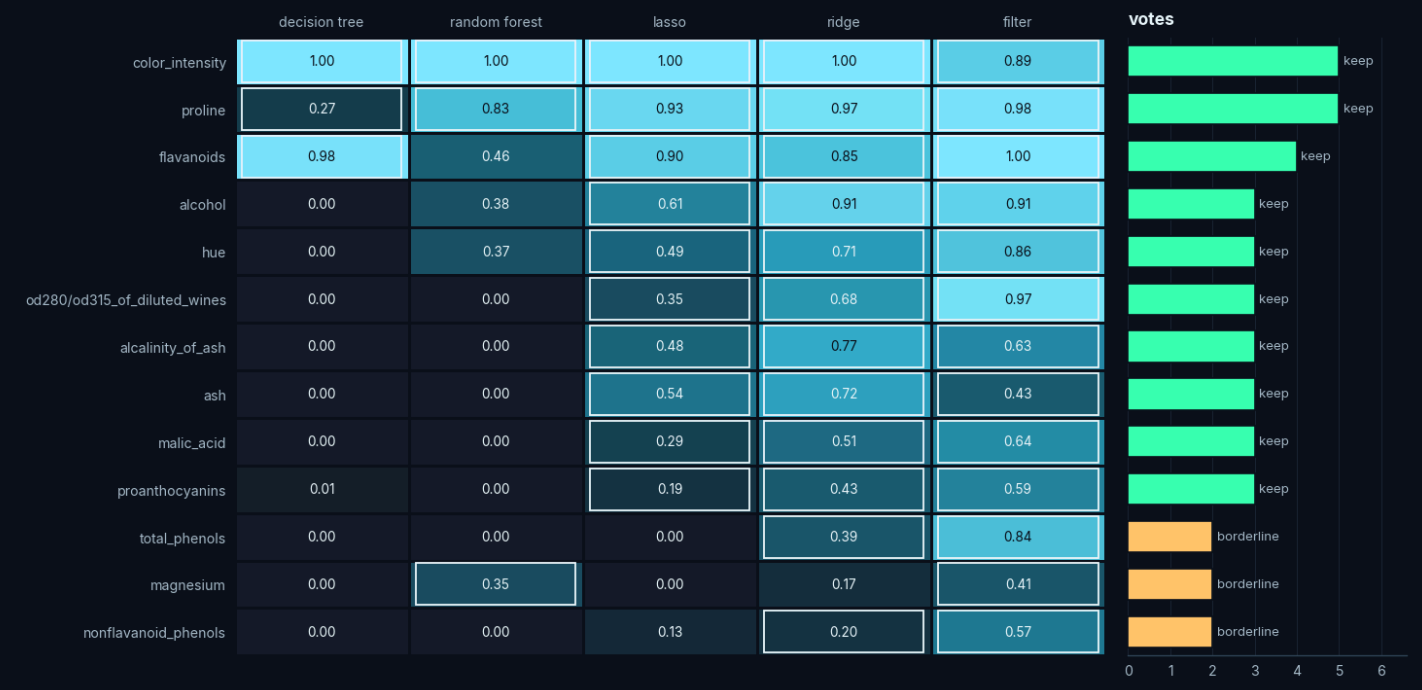
Keep: color_intensity, proline, flavanoids, alcohol, hue, od280/od315_of_diluted_wines, alcalinity_of_ash, ash, malic_acid, proanthocyanins · **Borderline:** total_phenols, magnesium, nonflavanoid_phenols · **Drop:** -

FEATURE	DECISION TREE	RANDOM FOREST	LASSO	RIDGE	FILTER	VOTES	DECISION
color_intensity	1.00 ✓	1.00 ✓	1.00 ✓	1.00 ✓	0.89 ✓	5	keep
proline	0.27 ✓	0.83 ✓	0.93 ✓	0.97 ✓	0.98 ✓	5	keep
flavanoids	0.98 ✓	0.46	0.90 ✓	0.85 ✓	1.00 ✓	4	keep
alcohol	0.00	0.38	0.61 ✓	0.91 ✓	0.91 ✓	3	keep
hue	0.00	0.37	0.49 ✓	0.71 ✓	0.86 ✓	3	keep
od280/od315_of_diluted_wines	0.00	0.00	0.35 ✓	0.68 ✓	0.97 ✓	3	keep
alcalinity_of_ash	0.00	0.00	0.48 ✓	0.77 ✓	0.63 ✓	3	keep
ash	0.00	0.00	0.54 ✓	0.72 ✓	0.43 ✓	3	keep
malic_acid	0.00	0.00	0.29 ✓	0.51 ✓	0.64 ✓	3	keep
proanthocyanins	0.01	0.00	0.19 ✓	0.43 ✓	0.59 ✓	3	keep
total_phenols	0.00	0.00	0.00	0.39 ✓	0.84 ✓	2	borderline
magnesium	0.00	0.35 ✓	0.00	0.17	0.41 ✓	2	borderline
nonflavanoid_phenols	0.00	0.00	0.13	0.20 ✓	0.57 ✓	2	borderline

Importances scaled to the strongest feature (1); ✓ = selected by that method. Decision tree depth 3 chosen by cross-validation; random forest: permutation importance on held-out rows (all one-hot columns of a feature shuffled together); lasso: non-zero coefficient at the penalty chosen by cross-validation; ridge: |coefficient| at least 10% of the largest; filter: significant univariate association.

cultivar · feature selection

FIGURE // 28_cultivar_selection.png



Cells: importance by each method, scaled to the strongest feature (1). Decision tree: impurity importance; random forest: drop in held-out score when the feature is shuffled; lasso / ridge: size of the standardised coefficient; filter: univariate association. Boxed = selected: above every shuffled shadow (tree, lasso, ridge), significant (forest, filter). Keep = selected by 3+ methods, borderline = 2.

05 / DISTRIBUTIONS

Distributions.

Normality is tested four ways (Shapiro-Wilk on up to 5,000 values, D'Agostino K^2, Jarque-Bera, Anderson-Darling); the verdict is 'not normal' when at least half of them reject at alpha 0.05. Every candidate distribution is fitted by maximum likelihood and ranked by AIC; the Akaike weight is its relative support among the candidates.

With thousands of values every test rejects even a good fit, so read KS D (the largest gap between the fitted and the observed cumulative distribution) as the size of the misfit: below 0.03 close, below 0.08 fair, above poor. KS p-values are optimistic since the parameters come from the same data. For integer columns the gap is measured on the integer support and the p-value comes from a chi-square goodness-of-fit test.

COLUMN	N	NORMALITY	TESTS REJECTING	BEST FIT	PARAMETERS	WEIGHT	ΔAIC TO 2ND	KS D	KS P	SHAPE
`alcohol`	178	not normal	3 of 4	normal	loc=13, scale=0.81	0.28	0.5 (chi-squared)	0.069 (fair)	0.36	roughly symmetric
`malic_acid`	178	not normal	4 of 4	lognormal	s=0.449, scale=2.11	0.99	10.8 (gamma)	0.130 (poor)	0.0043	heavily right-skewed
`ash`	178	not normal	3 of 4	logistic	loc=2.37, scale=0.151	0.66	2.0 (Student t)	0.048 (fair)	0.79	roughly symmetric, heavy-tailed
`alcalinity_of_ash`	178	normal	0 of 4	logistic	loc=19.4, scale=1.87	0.34	1.2 (normal)	0.047 (fair)	0.81	roughly symmetric
`magnesium`	178	not normal	4 of 4	lognormal	s=0.136, scale=98.8	0.88	5.5 (gamma)	0.070 (fair)	0.33	heavily right-skewed, heavy-tailed

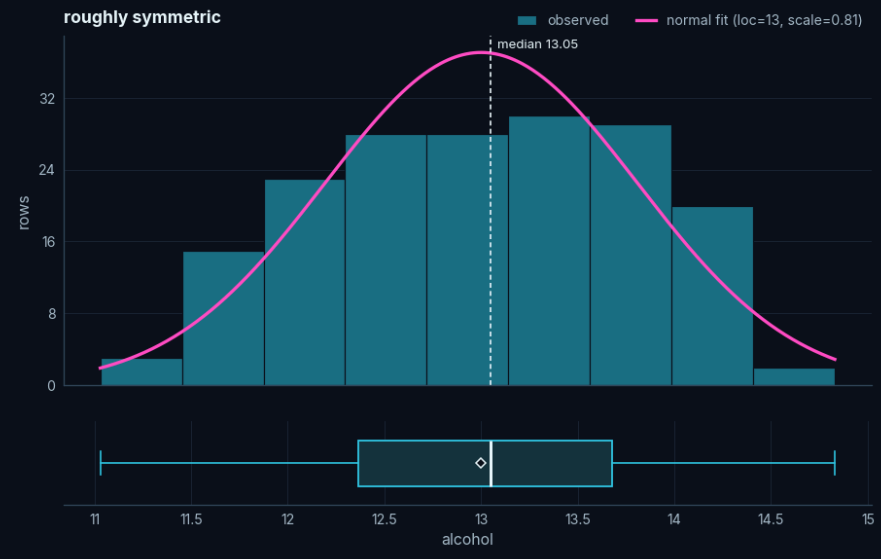
COLUMN	N	NORMALITY	TESTS REJECTING	BEST FIT	PARAMETERS	WEIGHT	ΔAIC TO 2ND	KS D	KS P	SHAPE
total_phenols	178	not normal	3 of 4	Weibull	c=4.12, scale=2.53	0.42	1.8 (normal)	0.072 (fair)	0.3	roughly symmetric
flavanoids	178	not normal	3 of 4	beta	a=1.25, b=2.27, loc=0.335, scale=4.75	0.55	0.4 (Weibull)	0.088 (poor)	0.12	roughly symmetric
nonflavanoid_phenols	178	not normal	4 of 4	gamma	a=8.42, scale=0.043	0.33	0.0 (chi-squared)	0.081 (poor)	0.19	roughly symmetric
proanthocyanins	178	not normal	3 of 4	gamma	a=7.26, scale=0.219	0.37	0.0 (chi-squared)	0.077 (fair)	0.23	moderately right-skewed
color_intensity	178	not normal	4 of 4	lognormal	s=0.459, scale=4.56	0.58	2.0 (gamma)	0.048 (fair)	0.78	moderately right-skewed
hue	178	not normal	2 of 4	normal	loc=0.957, scale=0.228	0.42	0.8 (Weibull)	0.063 (fair)	0.47	roughly symmetric
od280/od315_of_diluted_wines	178	not normal	4 of 4	beta	a=1.21, b=1.33, loc=1.27, scale=2.74	0.80	2.8 (uniform)	0.129 (poor)	0.0047	roughly symmetric, light-tailed
proline	178	not normal	4 of 4	lognormal	s=0.414, scale=686	0.68	2.5 (beta)	0.070 (fair)	0.32	moderately right-skewed

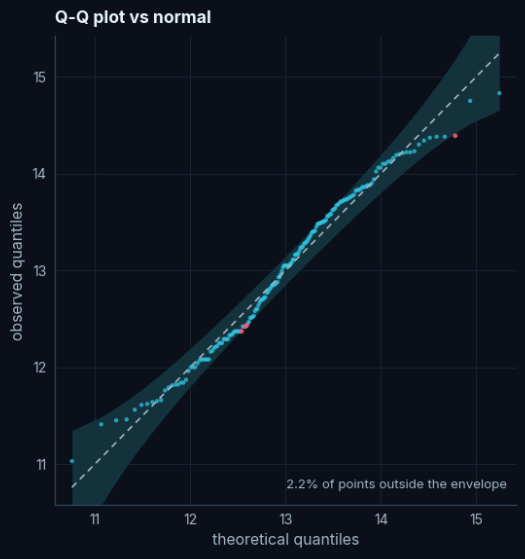
alcohol · distribution

FIGURE // 04_alcohol_dist.png

alcohol

numeric_continuous · n = 178 · mean 13 · median 13.05 · sd 0.8118 · skew -0.05 · not normal





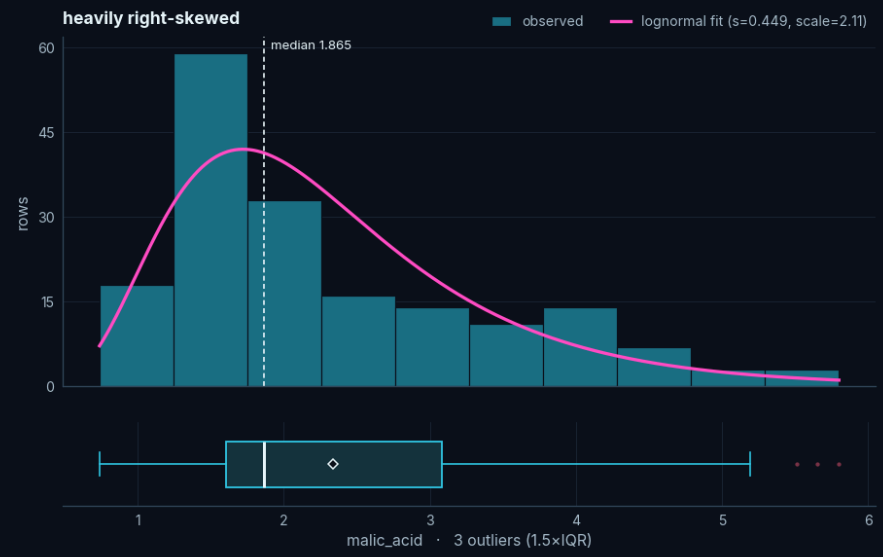
Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

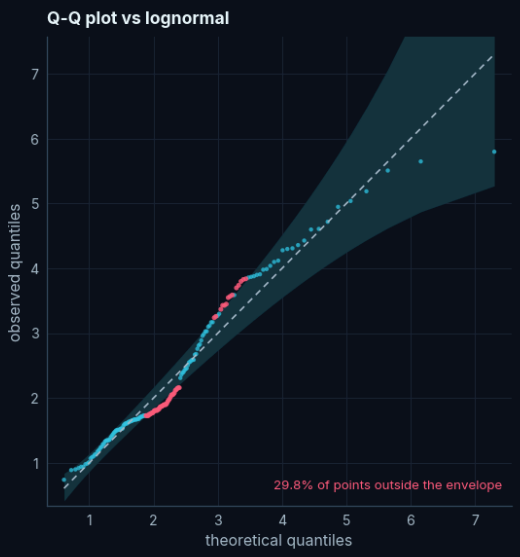
malic_acid · distribution

FIGURE // 05_malic_acid_dist.png

malic_acid

numeric_continuous · n = 178 · mean 2.336 · median 1.865 · sd 1.117 · skew 1.04 · not normal





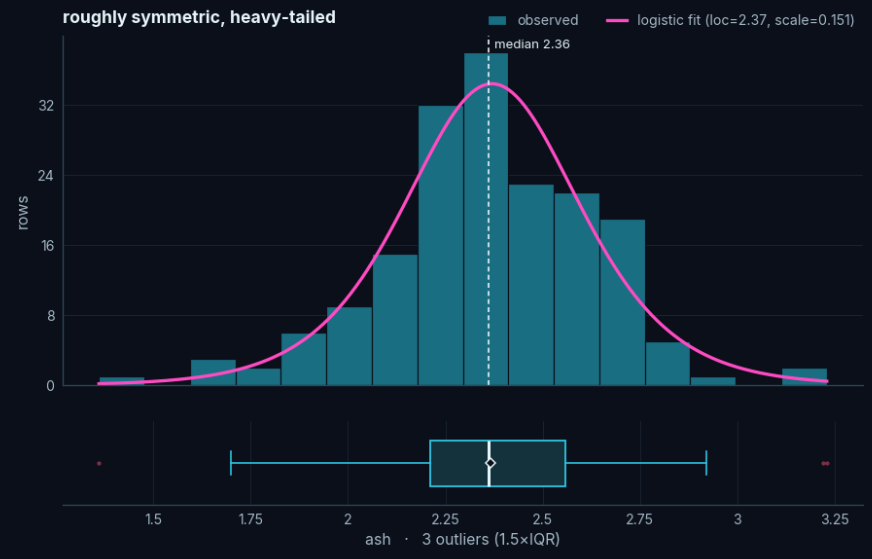
Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

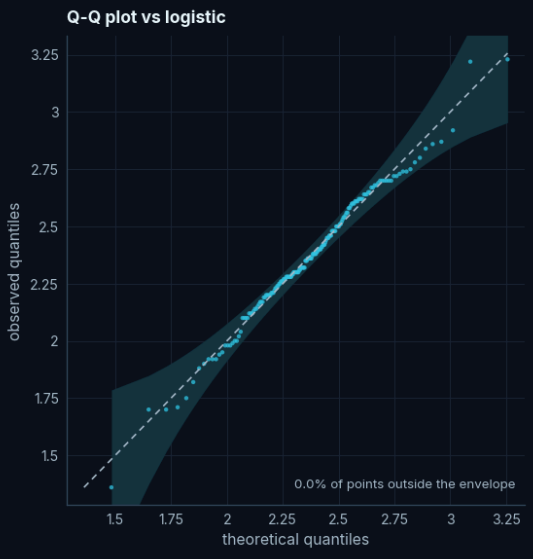
ash · distribution

FIGURE // 06_ash_dist.png

ash

numeric_continuous · n = 178 · mean 2.367 · median 2.36 · sd 0.2743 · skew -0.18 · not normal





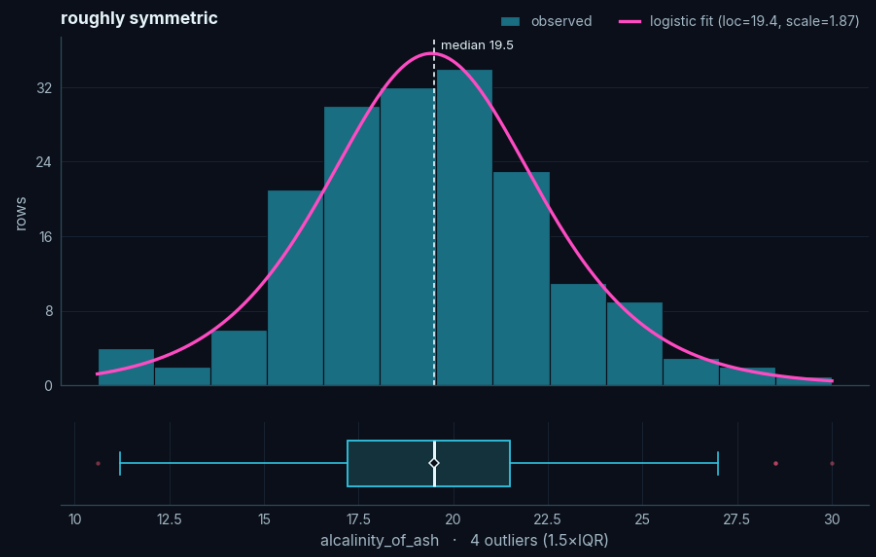
Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

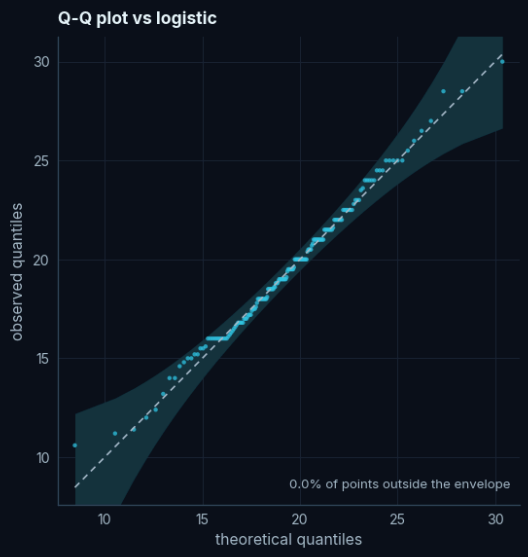
alcalinity_of_ash · distribution

FIGURE // 07_alcalinity_of_ash_dist.png

alcalinity_of_ash

numeric_continuous · n = 178 · mean 19.49 · median 19.5 · sd 3.34 · skew 0.21 · normal





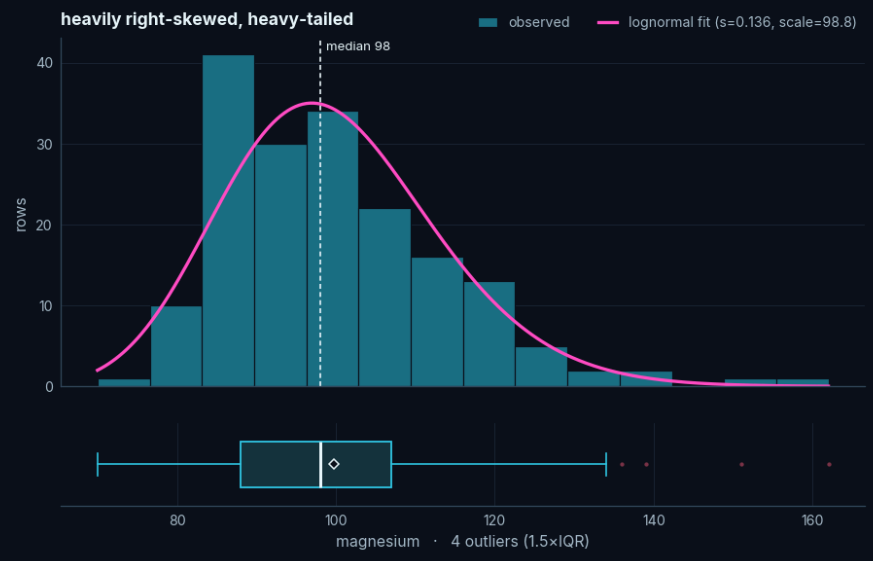
Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

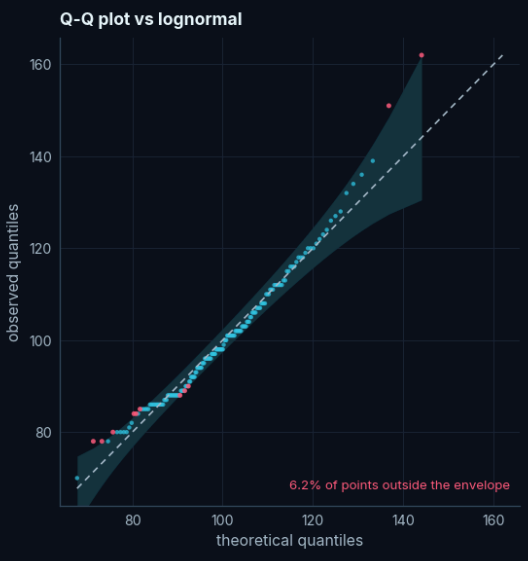
magnesium · distribution

FIGURE // 08_magnesium_dist.png

magnesium

numeric_continuous · n = 178 · mean 99.74 · median 98 · sd 14.28 · skew 1.10 · not normal





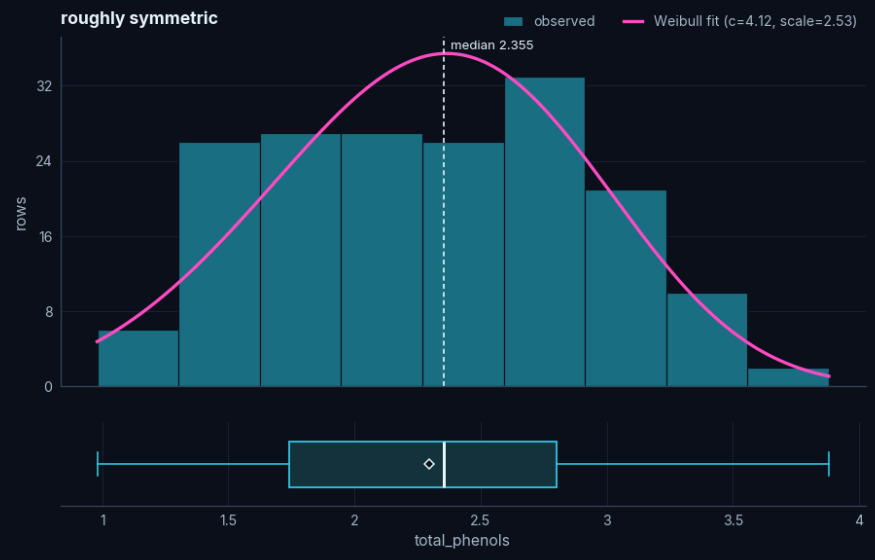
Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

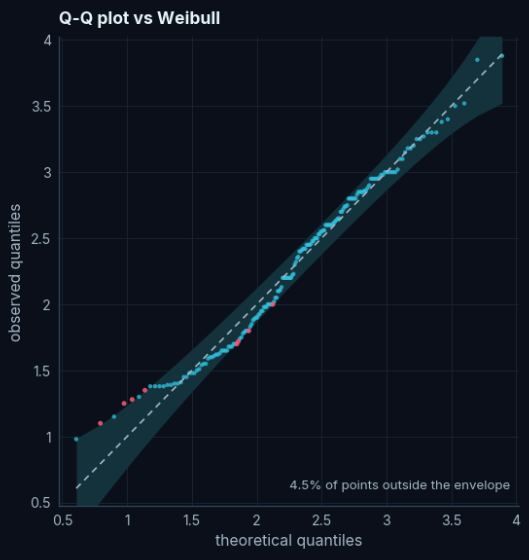
total_phenols · distribution

FIGURE // 09_total_phenols_dist.png

total_phenols

numeric_continuous · n = 178 · mean 2.295 · median 2.355 · sd 0.6259 · skew 0.09 · not normal





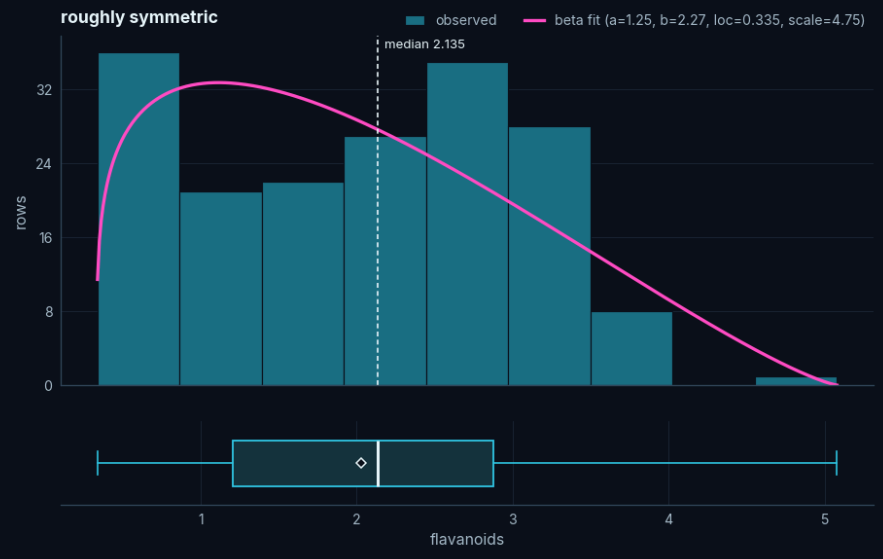
Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

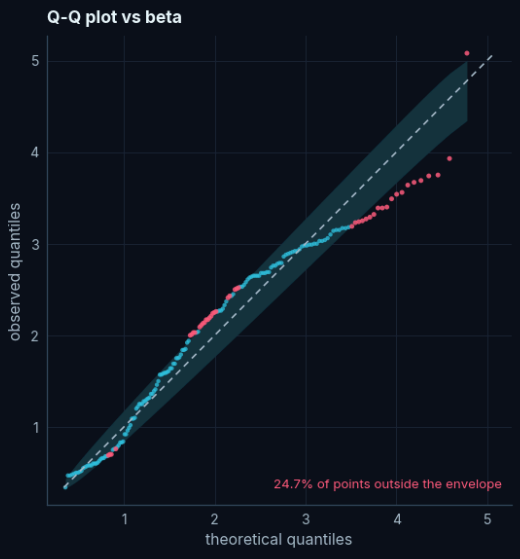
flavanoids · distribution

FIGURE // 10_flavanoids_dist.png

flavanoids

numeric_continuous · n = 178 · mean 2.029 · median 2.135 · sd 0.9989 · skew 0.03 · not normal





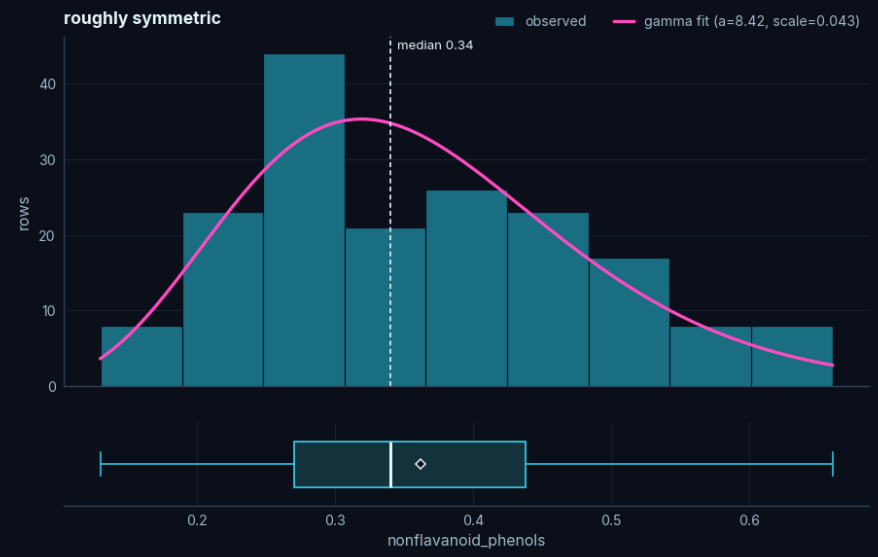
Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

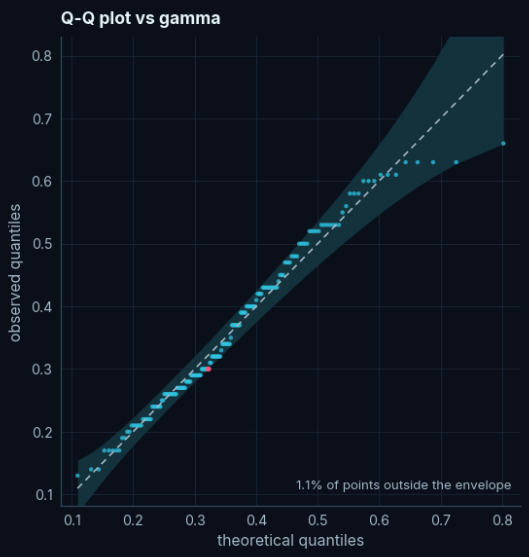
nonflavanoid_phenols · distribution

FIGURE // 11_nonflavanoid_phenols_dist.png

nonflavanoid_phenols

numeric_continuous · n = 178 · mean 0.3619 · median 0.34 · sd 0.1245 · skew 0.45 · not normal





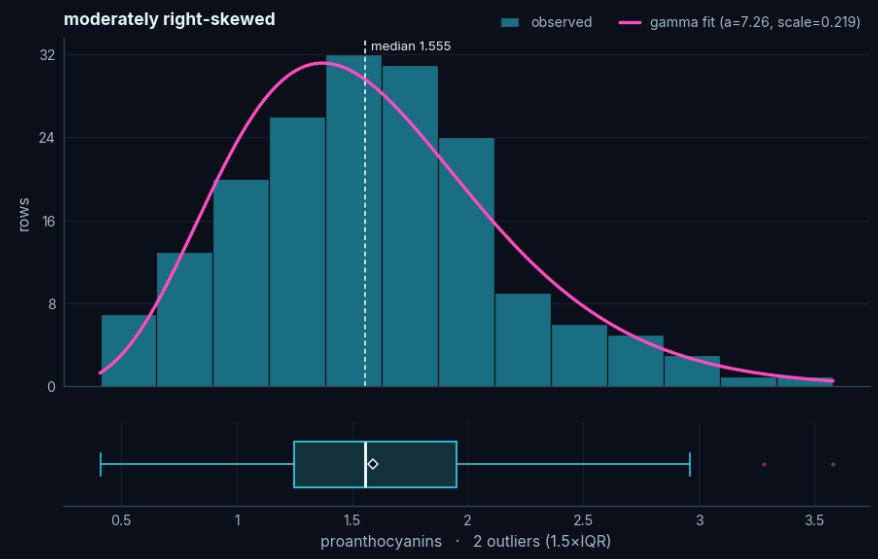
Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

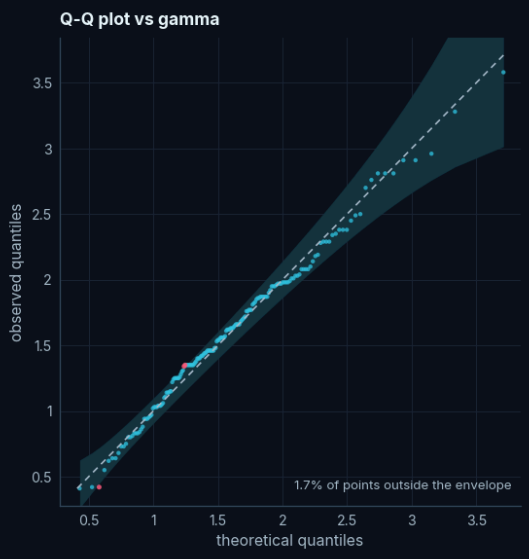
proanthocyanins · distribution

FIGURE // 12_proanthocyanins_dist.png

proanthocyanins

numeric_continuous · n = 178 · mean 1.591 · median 1.555 · sd 0.5724 · skew 0.52 · not normal





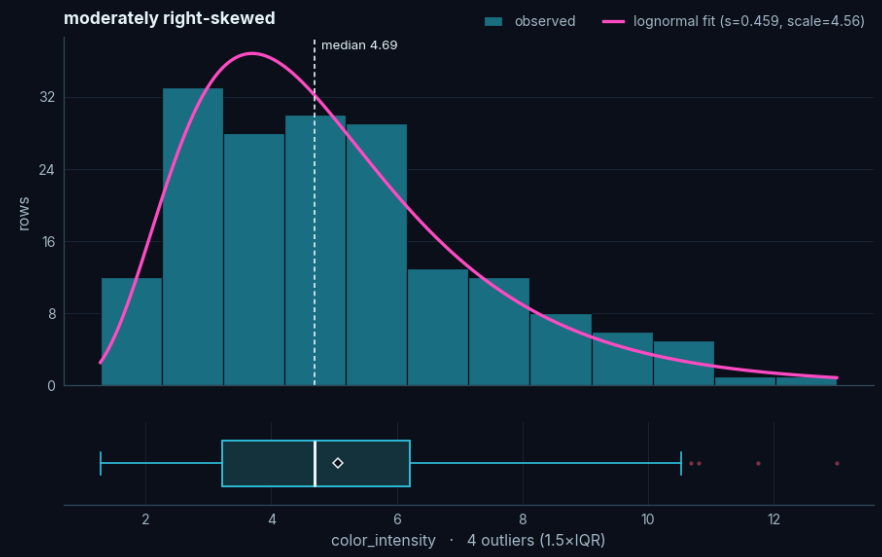
Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

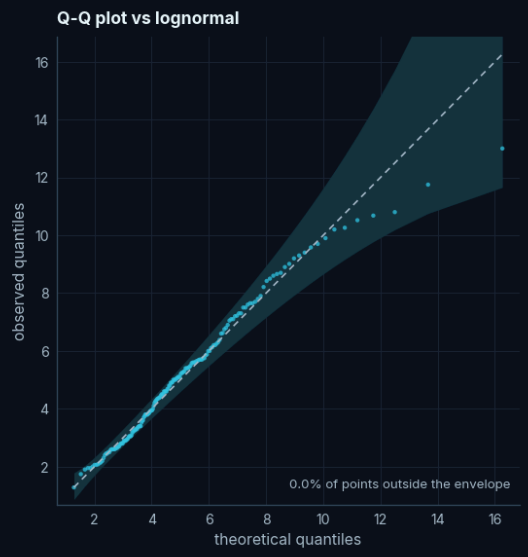
color_intensity · distribution

FIGURE // 13_color_intensity_dist.png

color_intensity

numeric_continuous · n = 178 · mean 5.058 · median 4.69 · sd 2.318 · skew 0.87 · not normal





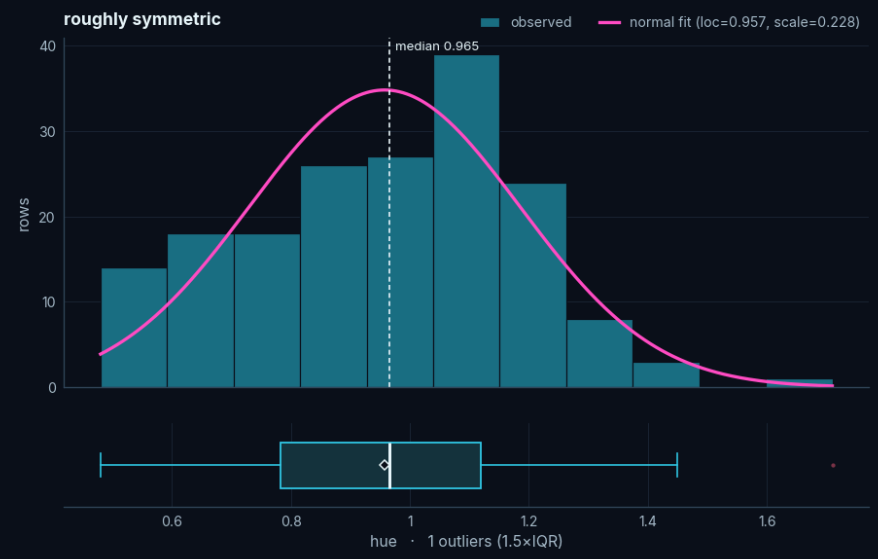
Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

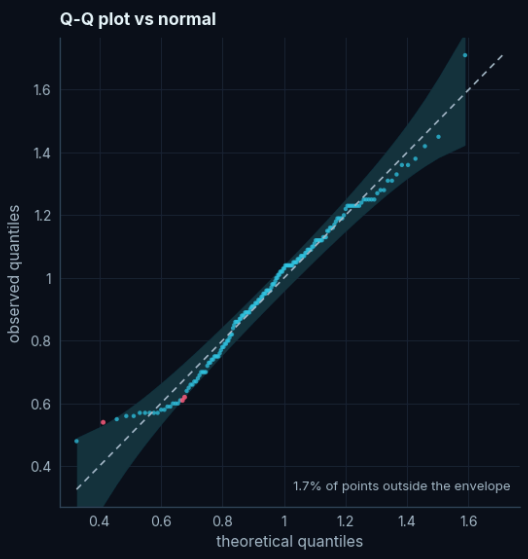
hue · distribution

FIGURE // 14_hue_dist.png

hue

numeric_continuous · n = 178 · mean 0.9574 · median 0.965 · sd 0.2286 · skew 0.02 · not normal





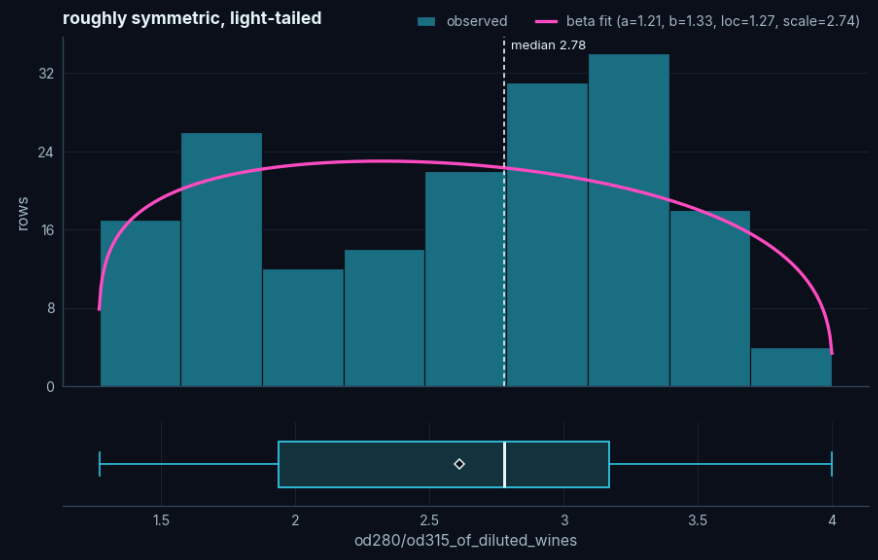
Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

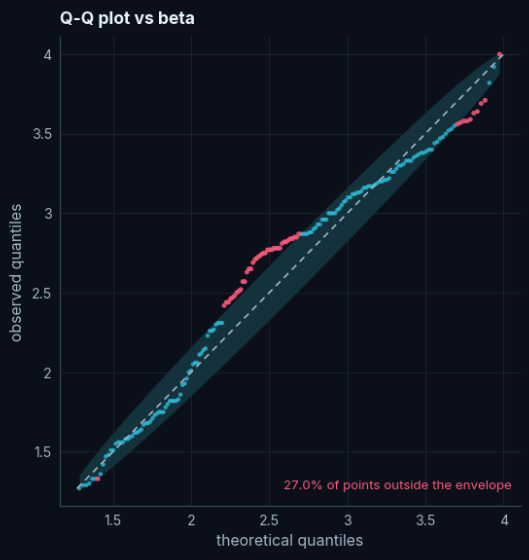
od280/od315_of_diluted_wines · distribution

FIGURE // 15_od280_od315_of_diluted_wines_dist.png

od280/od315_of_diluted_wines

numeric_continuous · n = 178 · mean 2.612 · median 2.78 · sd 0.71 · skew -0.31 · not normal





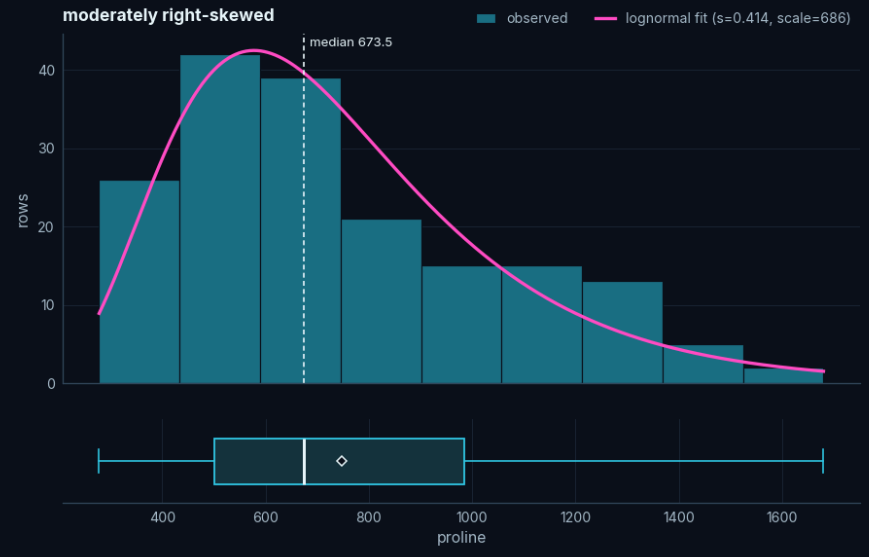
Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

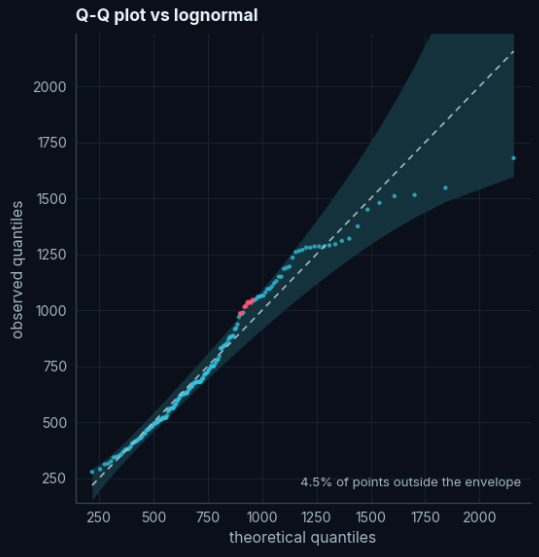
proline · distribution

FIGURE // 16_proline_dist.png

proline

numeric_continuous · n = 178 · mean 746.9 · median 673.5 · sd 314.9 · skew 0.77 · not normal





Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

06 / OUTLIERS

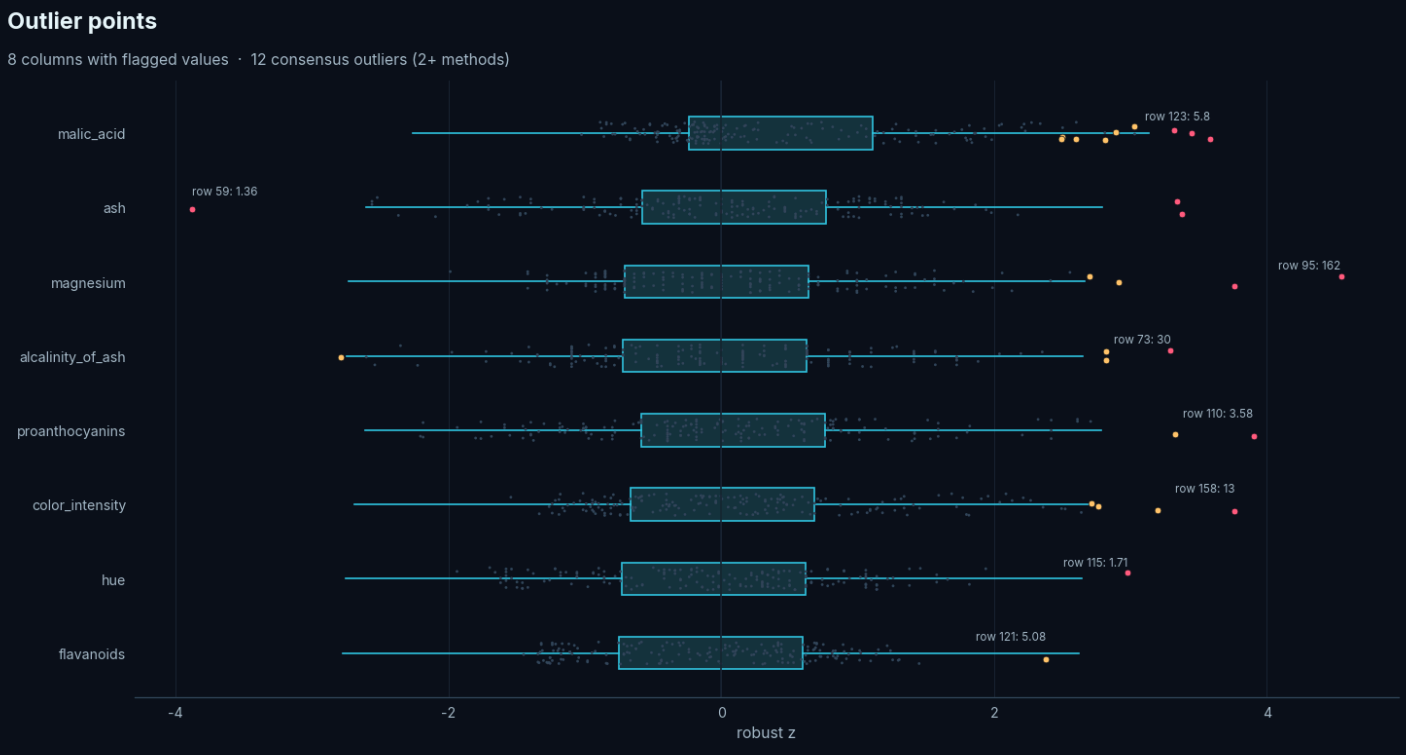
Outliers.

Three methods per numeric column: 3 sigma (|value - mean| > 3 standard deviations), boxplot (outside the Tukey fences Q1 - 1.5 IQR, Q3 + 1.5 IQR) and modified z (Hampel: |0.6745 (value - median) / MAD| > 3.5). A consensus outlier is flagged by at least two.

COLUMN	3 SIGMA	BOXPLOT	MODIFIED Z	CONSENSUS	FENCES	ACTION
alcohol	0	0	0	0 (0.0%)	[10.39, 15.65]	keep
malic_acid	1	3	9	3 (1.7%)	[-0.6175, 5.303]	reassign
ash	3	3	3	3 (1.7%)	[1.689, 3.079]	reassign

COLUMN	3 SIGMA	BOXPLOT	MODIFIED Z	CONSENSUS	FENCES	ACTION
alcalinity_of_ash	1	4	0	1 (0.6%)	[10.75, 27.95]	reassign
magnesium	2	4	2	2 (1.1%)	[59.5, 135.5]	reassign
total_phenols	0	0	0	0 (0.0%)	[0.1563, 4.386]	keep
flavanoids	1	0	0	0 (0.0%)	[-1.3, 5.38]	keep
nonflavanoid_phenols	0	0	0	0 (0.0%)	[0.01875, 0.6887]	keep
proanthocyanins	1	2	1	1 (0.6%)	[0.2, 3]	reassign
color_intensity	1	4	1	1 (0.6%)	[-1.25, 10.67]	reassign
hue	1	1	0	1 (0.6%)	[0.2762, 1.626]	reassign
od280/od315_of_diluted_wines	0	0	0	0 (0.0%)	[0.08875, 5.019]	keep
proline	0	0	0	0 (0.0%)	[-226.2, 1,711.8]	keep

Outlier points FIGURE // 29_dataset_outliers.png



Each column on a robust scale: (value − median) / (IQR / 1.349). Box = interquartile range, whiskers = boxplot fences. Red: flagged by 2-3 methods (3 sigma, boxplot, modified z); amber: by one. The most extreme points are labelled with their row.

Outlier points (12 consensus; the most extreme 40)

COLUMN	ROW	VALUE	Z	ROBUST Z	FLAGGED BY
magnesium	95	162	+4.36	+4.54	3 sigma, boxplot, modified z
proanthocyanins	110	3.58	+3.48	+3.90	3 sigma, boxplot, modified z
ash	59	1.36	-3.67	-3.88	3 sigma, boxplot, modified z
magnesium	69	151	+3.59	+3.76	3 sigma, boxplot, modified z
color_intensity	158	13	+3.43	+3.76	3 sigma, boxplot, modified z

COLUMN	ROW	VALUE	Z	ROBUST Z	FLAGGED BY
malic_acid	123	5.8	+3.10	+3.59	3 sigma, boxplot, modified z
malic_acid	173	5.65	+2.97	+3.45	boxplot, modified z
ash	121	3.23	+3.15	+3.38	3 sigma, boxplot, modified z
ash	25	3.22	+3.11	+3.34	3 sigma, boxplot, modified z
malic_acid	137	5.51	+2.84	+3.32	boxplot, modified z
alcalinity_of_ash	73	30	+3.15	+3.29	3 sigma, boxplot
hue	115	1.71	+3.29	+2.98	3 sigma, boxplot

Every flagged point, with the methods that flag it: `outliers.csv`

Rows that are outliers on several columns at once

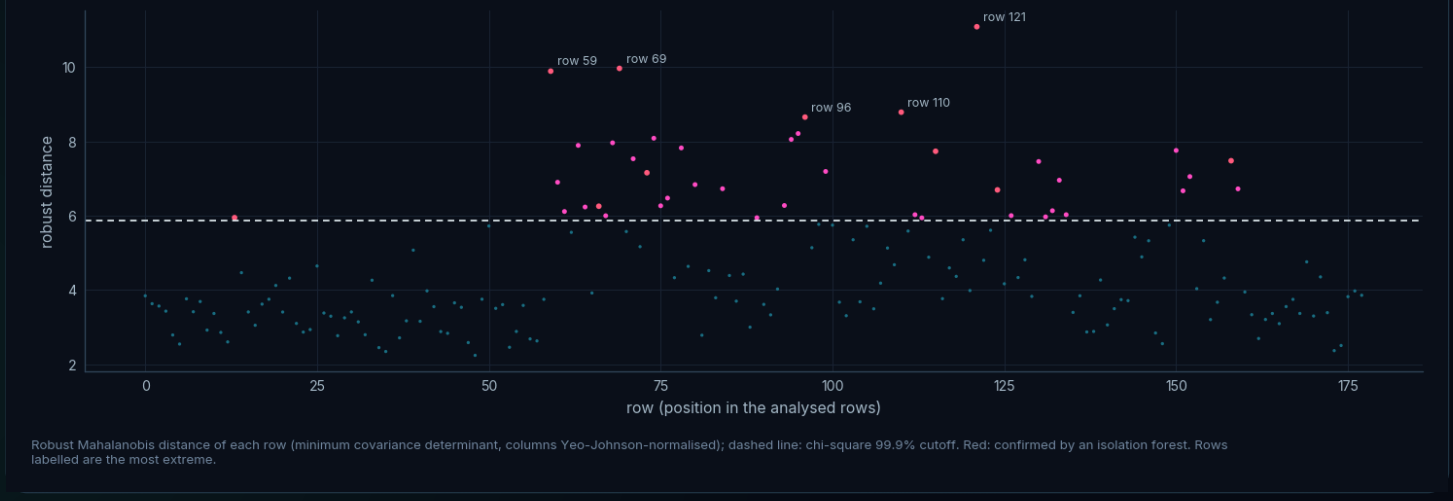
Robust Mahalanobis distance (minimum covariance determinant) over 13 numeric columns: 41 rows (23.0%) beyond the chi-square 99.9% cutoff (5.88); 11 also isolated by an isolation forest.

> **Caution:** 23% of rows beyond the cutoff: the columns are far from elliptical (clusters, heavy tails), so read only the rows confirmed by both methods.

ROW	ROBUST DISTANCE	ISOLATION FOREST	VALUES DRIVING IT
121	11.09	yes	ash=3.23, alcalinity_of_ash=28.5, flavanoids=5.08
69	9.96	yes	magnesium=151, ash=1.75, proanthocyanins=2.5
59	9.89	yes	ash=1.36, alcalinity_of_ash=10.6, proanthocyanins=0.42
110	8.79	yes	proanthocyanins=3.58, ash=1.82, malic_acid=3.74
96	8.65	yes	magnesium=134, nonflavanoid_phenols=0.14, ash=2.74
95	8.21	no	magnesium=162, proanthocyanins=3.28, color_intensity=2.6
74	8.08	no	nonflavanoid_phenols=0.13, total_phenols=3.38, alcohol=11.96
94	8.05	no	alcohol=11.62, nonflavanoid_phenols=0.17, proline=345
68	7.96	no	proanthocyanins=0.42, nonflavanoid_phenols=0.55, od280/od315_of_diluted_wines=1.93
63	7.89	no	total_phenols=3.5, nonflavanoid_phenols=0.19, hue=1.22
78	7.82	no	magnesium=136, proanthocyanins=2.76, ash=1.95
150	7.76	no	magnesium=123, color_intensity=8.6, od280/od315_of_diluted_wines=1.3
115	7.73	yes	hue=1.71, alcohol=11.03, nonflavanoid_phenols=0.52
71	7.53	no	alcalinity_of_ash=25, hue=1.36, ash=2.67
158	7.48	yes	color_intensity=13, proanthocyanins=2.7, alcalinity_of_ash=25
130	7.46	no	magnesium=122, od280/od315_of_diluted_wines=1.29, proanthocyanins=0.94
99	7.19	no	proanthocyanins=2.81, hue=1.42, malic_acid=3.17
73	7.15	yes	alcalinity_of_ash=30, magnesium=139, hue=1.31
152	7.05	no	alcalinity_of_ash=25.5, od280/od315_of_diluted_wines=1.33, ash=2.75

ROW	ROBUST DISTANCE	ISOLATION FOREST	VALUES DRIVING IT
133	6.95	no	od280/od315_of_diluted_wines=1.29, malic_acid=3.55, proanthocyanins=0.84

Multivariate outliers FIGURE // 30_dataset_multivariate_outliers.png



Robust Mahalanobis distance of each row (minimum covariance determinant, columns Yeo-Johnson-normalised); dashed line: chi-square 99.9% cutoff. Red: confirmed by an isolation forest. Rows labelled are the most extreme.

07 / DATA QUALITY

Data quality.

SEVERITY	COLUMN	MESSAGE
medium	-	rows are ordered (sorted file or time series): lag-1 autocorrelation flavanoids +0.79, od280/od315_of_diluted_wines +0.76, color_intensity +0.74, proline +0.74, hue +0.69, alcohol +0.65 (+6 more) → shuffle before a random split, or use time-aware validation
info	-	dtype optimizations save ~9.0 KB: alcohol → float32, malic_acid → float32, ash → float32, alcalinity_of_ash → float32, magnesium → float32, total_phenols → float32, flavanoids → float32, nonflavanoid_phenols → float32 (+5 more) (see quality['dtype_optimizations'] for code)
info	-	PCA recommended: 3 components (parallel analysis) keep 67% of the variance of 13 columns (KMO 0.78 middling, Bartlett p 2e-224)
info	-	factor analysis: 3 factors explain 70% of the variance of 12 variables (KMO 0.79, Bartlett p 4e-220); 3 reliable scales created
info	-	3 columns show heaping (excess of values ending in 0 or 5): alcalinity_of_ash, total_phenols, proline → details per column

Memory: suggested dtypes save 9.0 KB

COLUMN	CURRENT	SUGGESTED	SAVING	CODE
`alcohol`	float64	float32	712 B (50%)	`df['alcohol'] = df['alcohol'].astype('float32')`
`malic_acid`	float64	float32	712 B (50%)	`df['malic_acid'] = df['malic_acid'].astype('float32')`
`ash`	float64	float32	712 B (50%)	`df['ash'] = df['ash'].astype('float32')`
`alcalinity_of_ash`	float64	float32	712 B (50%)	`df['alcalinity_of_ash'] = df['alcalinity_of_ash'].astype('float32')`
`magnesium`	float64	float32	712 B (50%)	`df['magnesium'] = df['magnesium'].astype('float32')`

COLUMN	CURRENT	SUGGESTED	SAVING	CODE
`total_phenols`	float64	float32	712 B (50%)	`df['total_phenols'] = df['total_phenols'].astype('float32')`
`flavanoids`	float64	float32	712 B (50%)	`df['flavanoids'] = df['flavanoids'].astype('float32')`
`nonflavanoid_phenols`	float64	float32	712 B (50%)	`df['nonflavanoid_phenols'] = df['nonflavanoid_phenols'].astype('float32')`
`proanthocyanins`	float64	float32	712 B (50%)	`df['proanthocyanins'] = df['proanthocyanins'].astype('float32')`
`color_intensity`	float64	float32	712 B (50%)	`df['color_intensity'] = df['color_intensity'].astype('float32')`
`hue`	float64	float32	712 B (50%)	`df['hue'] = df['hue'].astype('float32')`
`od280/od315_of_diluted_wines`	float64	float32	712 B (50%)	`df['od280/od315_of_diluted_wines'] = df['od280/od315_of_diluted_wines'].astype('float32')`
`proline`	float64	float32	712 B (50%)	`df['proline'] = df['proline'].astype('float32')`

08 / MULTIVARIATE STATISTICS

Multivariate statistics.

Dimensionality reduction

Numeric columns	13 (standardised; target, identifiers and constants left out)
Bartlett sphericity	chi² 1,317.2, p 2e-224 (significant = the columns share information)
KMO adequacy	0.78 (middling)
Components to keep	3 by parallel analysis, 3 by Kaiser (eigenvalue > 1), 5 / 8 / 10 for 80 / 90 / 95% of the variance
PCA	**recommended**: 3 components (parallel analysis) keep 67% of the variance of 13 columns (KMO 0.78 middling, Bartlett p 2e-224)
Factor analysis	**recommended**: 3 factors explain 70% of the variance of 12 variables (KMO 0.79, Bartlett p 4e-220); 3 reliable scales created

- **PC1** (36.2%): flavanoids, total_phenols, od280/od315_of_diluted_wines, proanthocyanins
- **PC2** (19.2%): color_intensity, alcohol, proline, ash
- **PC3** (11.1%): ash, alcalinity_of_ash, alcohol, nonflavanoid_phenols

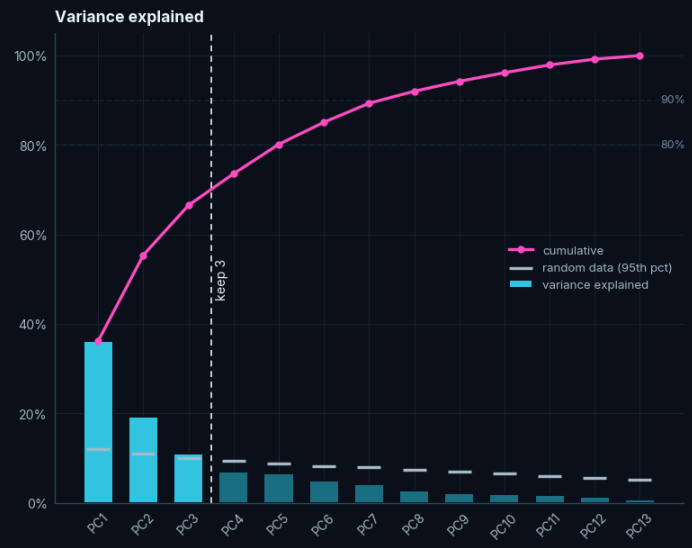
```
from sklearn.pipeline import make_pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
cols = ['alcohol', 'malic_acid', 'ash', 'alcalinity_of_ash', 'magnesium', 'total_phenols', 'flavanoids',
'nonflavanoid_phenols', 'proanthocyanins', 'color_intensity', 'hue', 'od280/od315_of_diluted_wines', 'proline']
pca = make_pipeline(SimpleImputer(strategy='median'), StandardScaler(), PCA(n_components=3))
scores = pca.fit_transform(df[cols])
```

Principal components

FIGURE // 31_dataset_pca.png

Principal components · PCA recommended

13 numeric columns · keep 3 components = 67% of the variance · KMO 0.78 (middling) · Bartlett p 2e-224



Loadings (correlation with the component)

	PC1	PC2	PC3
flavanoids	**0.92**	-0.01	0.18
ash	-0.00	0.50	**0.75**
alcalinity_of_ash	**-0.52**	-0.02	**0.74**
total_phenols	**0.86**	0.10	0.18
od280/od315_of_diluted_wines	**0.82**	-0.26	0.20
color_intensity	-0.19	**0.84**	-0.17
alcohol	0.31	**0.76**	-0.25
proline	**0.62**	**0.58**	-0.15
hue	**0.64**	-0.44	0.10
proanthocyanins	**0.68**	0.06	0.18
nonflavanoid_phenols	**-0.65**	0.05	0.20
malic_acid	**-0.53**	0.36	0.11
magnesium	0.31	0.47	0.16

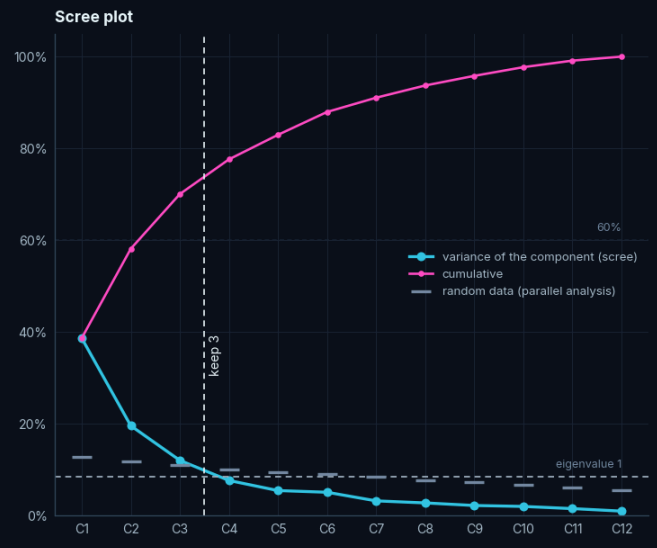
Bars: variance explained by each component of the standardised columns; line: cumulative. Grey ticks: what random data of the same size would give (parallel analysis): components above them carry real structure. Right: correlation of each column with each kept component.

Factor analysis

FIGURE // 32_dataset_factors.png

Factor analysis (principal components, varimax)

3 factors · KMO 0.79 · Bartlett p 4e-220 · F1 α 0.87 · F2 α 0.75 · F3 SB 0.61



Rotated component matrix

	F1	F2	F3
flavanoids	**0.93**	0.17	-0.03
od280/od315_of_diluted_wines	**0.87**	-0.10	-0.07
total_phenols	**0.86**	0.26	0.02
hue	**0.70**	-0.31	-0.20
proanthocyanins	**0.69**	0.16	0.03
nonflavanoid_phenols	**-0.55**	-0.13	0.38
malic_acid	**-0.52**	0.20	0.35
alcohol	0.12	**0.87**	-0.04
color_intensity	-0.34	**0.80**	0.17
proline	**0.46**	**0.72**	-0.10
ash	0.14	0.26	**0.85**
alcalinity_of_ash	-0.28	-0.33	**0.79**

Left: share of the variance of each component (scree plot); a component is kept while it beats random data of the same size (grey ticks, parallel analysis). The Kaiser criterion (eigenvalue 1, dashed) and the cumulative share against 60% are shown for reference. Right: rotated loadings, items grouped by the factor they load on most; bold when |loading| ≥ 0.4. Removed before this solution: magnesium.

Factor analysis

Principal components of the correlation matrix, varimax rotation, then Cronbach's alpha per factor; items are examined and removed one at a time (communality < .40, highest loading < .40 or cross-loading), and the analysis is run again after each removal.

STEP	QUESTION	ANSWER	
1	Quantitative or Likert variables?	13 usable: alcohol (quantitative), malic_acid (quantitative), ash (quantitative), alcalinity_of_ash (quantitative), magnesium (quantitative), total_phenols (quantitative), flavanoids (quantitative), nonflavanoid_phenols (quantitative), proanthocyanins (quantitative), color_intensity (quantitative), hue (quantitative), od280/od315_of_diluted_wines (quantitative), proline (quantitative)	YES
1	Missing values correctly declared?	yes: no missing-value code among the items	YES

STEP	QUESTION	ANSWER	
1	Determinant of the correlation matrix	4.69e-04 (> 0.00001: no singularity)	YES
1	Variables on the same scale?	no: items standardised (z-scores) for the reliability analysis and the scores; extraction uses the correlation matrix either way	YES
1	Sample size sufficient?	178 complete rows (listwise) for 13 variables: 14 per variable (rule: at least 100 and about 10 per variable)	YES
2	Extraction	principal components of the correlation matrix, eigenvalues shown with the scree plot against those of random data (parallel analysis); varimax rotation (Kaiser normalisation); loadings sorted by size	DONE
3	KMO > .50?	KMO = 0.78 on the 13 variables	YES
3	Bartlett Sig. < .05?	chi2 = 1,317.2, p = 2e-224	YES
3	KMO and Bartlett on the retained items	KMO = 0.79, Bartlett p = 4e-220 on 12 items	YES
4	Examine magnesium	communality 0.34 < 0.40: poorly explained by the factors → removed, analysis run again	REMOVED
4	Communalities (extraction) all >= .40?	yes: lowest 0.43 (malic_acid)	YES
5	How many factors?	parallel analysis (Horn): 3 → 3 (70% of the variance); eigenvalue > 1: 3; cumulative variance >= 60%: 3; scree elbow: 3 (all agree)	YES
6	Rotated component matrix	F1 = flavanoids, od280/od315_of_diluted_wines, total_phenols, hue, proanthocyanins, nonflavanoid_phenols, malic_acid; F2 = alcohol, color_intensity, proline; F3 = ash, alcalinity_of_ash	YES
7	Reliability of F1	alpha = 0.87 (very good) on 7 items; reversed: nonflavanoid_phenols, malic_acid	YES
7	Reliability of F2	alpha = 0.75 (good) on 3 items	YES
7	Reliability of F3	two items: Spearman-Brown = 0.61 (acceptable; r = 0.44)	YES
8	Create the factors	F1_flavanoids_od280/od315_of_diluted_wines, F2_alcohol_color_intensity, F3_ash_alcalinity_of_ash (mean of the items)	YES
9	Descriptives of the new factors	F1_flavanoids_od280/od315_of_diluted_wines: mean -0.00, median 0.19, mode -1.09, sd 0.75; F2_alcohol_color_intensity: mean -0.00, median -0.02, mode 0.45, sd 0.82; F3_ash_alcalinity_of_ash: mean -0.00, median -0.01, mode -0.68, sd 0.85	DONE

Component matrices (sorted by size; each item goes to its highest rotated loading; communality = share of the item explained)

ITEM	F1 UNROTATED	F2 UNROTATED	F3 UNROTATED	F1 ROTATED	F2 ROTATED	F3 ROTATED	COMMUNALITY	FACTOR
flavanoids	+0.92	+0.05	+0.20	+0.93	+0.17	-0.03	0.89	F1
od280/od315_of_diluted_wines	+0.83	-0.21	+0.21	+0.87	-0.10	-0.07	0.78	F1
total_phenols	+0.85	+0.16	+0.21	+0.86	+0.26	+0.02	0.80	F1
hue	+0.66	-0.42	+0.08	+0.70	-0.31	-0.20	0.62	F1
proanthocyanins	+0.67	+0.09	+0.19	+0.69	+0.16	+0.03	0.50	F1
nonflavanoid_phenols	-0.64	+0.03	+0.22	-0.55	-0.13	+0.38	0.46	F1

ITEM	F1 UNROTATED	F2 UNROTATED	F3 UNROTATED	F1 ROTATED	F2 ROTATED	F3 ROTATED	COMMUNALITY	FACTOR
malic_acid	-0.55	+0.34	+0.13	-0.52	+0.20	+0.35	0.43	F1
alcohol	+0.28	+0.81	-0.19	+0.12	+0.87	-0.04	0.78	F2
color_intensity	-0.23	+0.85	-0.12	-0.34	+0.80	+0.17	0.78	F2
proline	+0.59	+0.61	-0.12	+0.46	+0.72	-0.10	0.74	F2
ash	-0.04	+0.46	+0.77	+0.14	+0.26	+0.85	0.81	F3
alcalinity_of_ash	-0.52	-0.08	+0.73	-0.28	-0.33	+0.79	0.81	F3

Reliability: Cronbach's alpha per factor (excellent >= .90, very good >= .80, good >= .70, acceptable >= .60, poor >= .50). (+) marks a deletion that would raise alpha by .05 or more: delete only if it also makes sense for the construct; marginal gains are not worth an item. A two-item factor is judged on the Spearman-Brown coefficient 2r / (1 + r), which alpha understates for two items.

FACTOR	ITEMS	RELIABILITY		ALPHA IF ITEM DELETED
F1	flavanoids, od280/od315_of_diluted_wines, total_phenols, hue, proanthocyanins, nonflavanoid_phenols (reversed), malic_acid (reversed)	0.87	very good	flavanoids: 0.82, od280/od315_of_diluted_wines: 0.83, total_phenols: 0.84, hue: 0.86, proanthocyanins: 0.86, nonflavanoid_phenols: 0.87, malic_acid: 0.88
F2	alcohol, color_intensity, proline	0.75	good	alcohol: 0.48, color_intensity: 0.78, proline: 0.71
F3	ash, alcalinity_of_ash	0.61 (Spearman-Brown; r 0.44, alpha 0.61)	acceptable	-

New factors (created only for factors with an acceptable alpha: the mean of their items)

FACTOR	ITEMS	N	MEAN	MEDIAN	MODE	STD
F1_flavanoids_od280/od315_of_diluted_wines	flavanoids, od280/od315_of_diluted_wines, total_phenols, hue, proanthocyanins, nonflavanoid_phenols, malic_acid	178	-0.00	0.19	-1.09	0.75
F2_alcohol_color_intensity	alcohol, color_intensity, proline	178	-0.00	-0.02	0.45	0.82
F3_ash_alcalinity_of_ash	ash, alcalinity_of_ash	178	-0.00	-0.01	-0.68	0.85

```
items = df[['flavanoids', 'od280/od315_of_diluted_wines', 'total_phenols', 'hue', 'proanthocyanins',
'nonflavanoid_phenols', 'malic_acid']].copy()
items = (items - items.mean()) / items.std()  # different scales: z-scores
items['nonflavanoid_phenols'] = -items['nonflavanoid_phenols']  # reversed item (negative loading)
items['malic_acid'] = -items['malic_acid']  # reversed item (negative loading)
df['F1_flavanoids_od280/od315_of_diluted_wines'] = items.mean(axis=1)  # MEAN: the items answered
# MEAN.7 (every item required): items.mean(axis=1).where(items.notna().all(axis=1))

items = df[['alcohol', 'color_intensity', 'proline']].copy()
items = (items - items.mean()) / items.std()  # different scales: z-scores
df['F2_alcohol_color_intensity'] = items.mean(axis=1)  # MEAN: the items answered
# MEAN.3 (every item required): items.mean(axis=1).where(items.notna().all(axis=1))

items = df[['ash', 'alcalinity_of_ash']].copy()
items = (items - items.mean()) / items.std()  # different scales: z-scores
df['F3_ash_alcalinity_of_ash'] = items.mean(axis=1)  # MEAN: the items answered
# MEAN.2 (every item required): items.mean(axis=1).where(items.notna().all(axis=1))
```

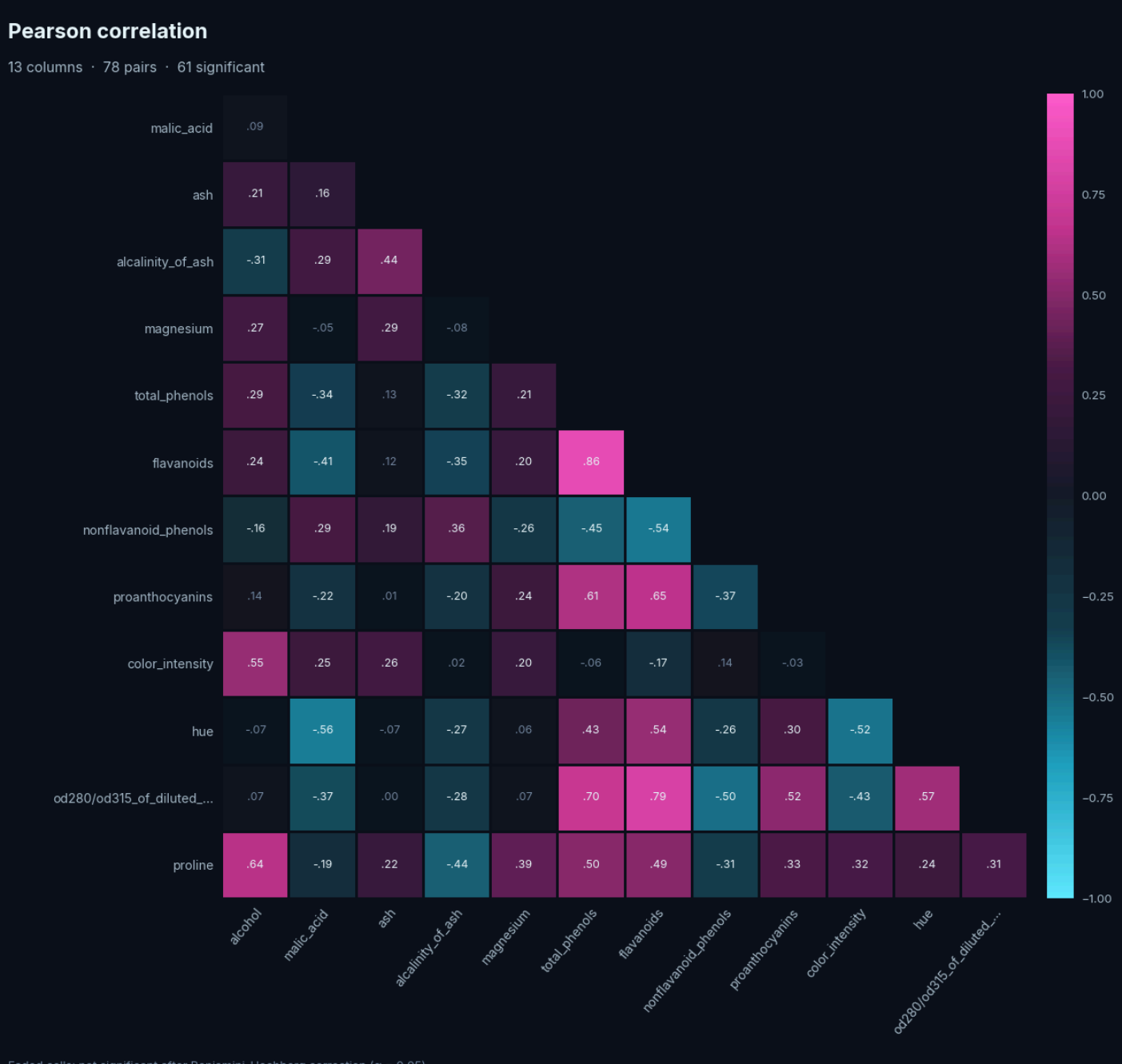
Groups of redundant columns

COLUMNS	MEAN |SPEARMAN|	KEEP
total_phenols, flavanoids, od280/od315_of_diluted_wines	0.77	`flavanoids`

Associations between columns

91 pairs tested, 74 significant after Benjamini-Hochberg correction (alpha 0.05). Two numeric columns are tested with Pearson when both are roughly symmetric with light tails, with Spearman otherwise: the choice never depends on which coefficient is the larger.

Pearson correlation FIGURE // 01_dataset_corr_pearson.png



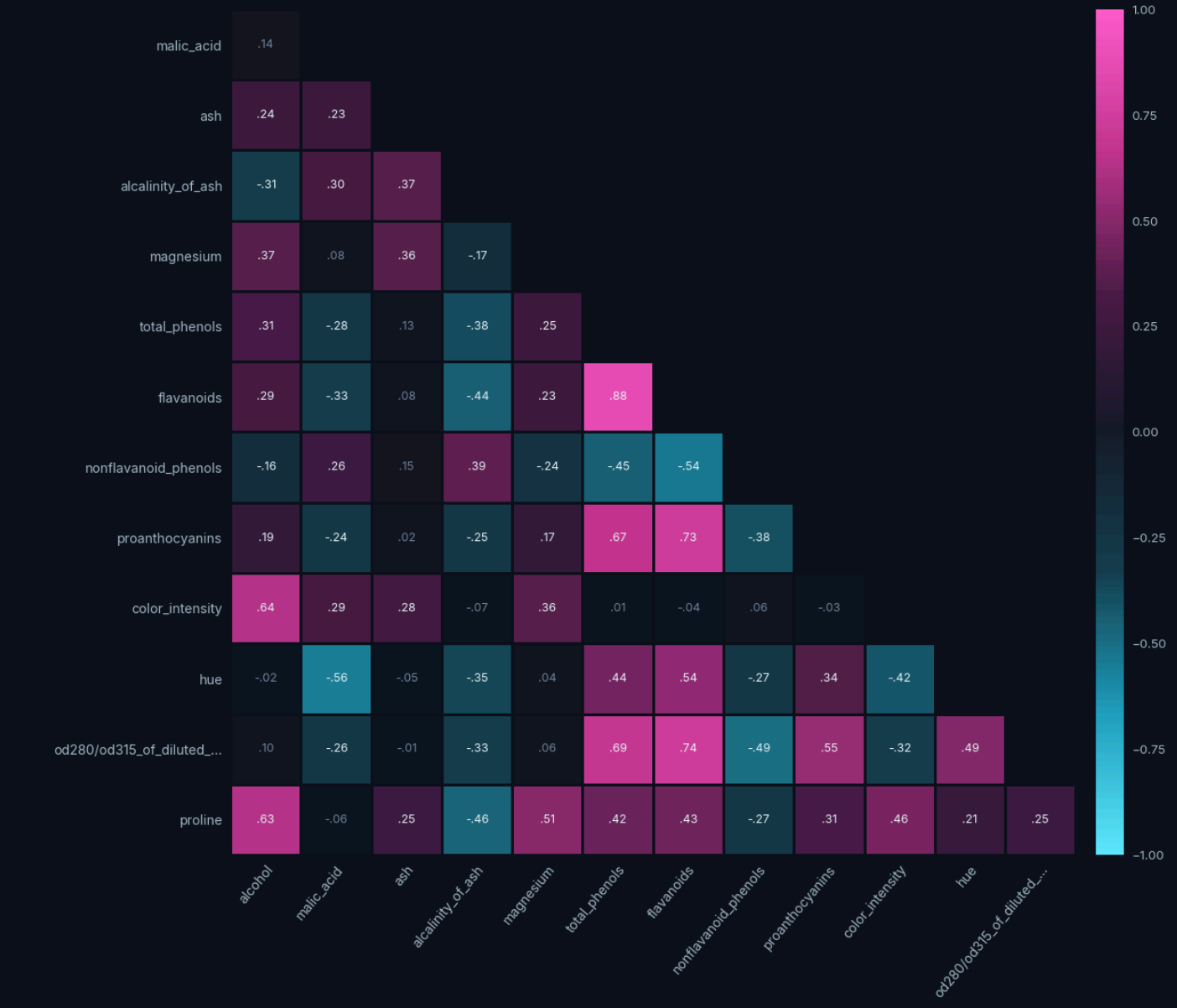
Faded cells: not significant after Benjamini-Hochberg correction (α = 0.05).

Spearman correlation

FIGURE // 02_dataset_corr_spearman.png

Spearman rank correlation

13 columns · 78 pairs · 60 significant



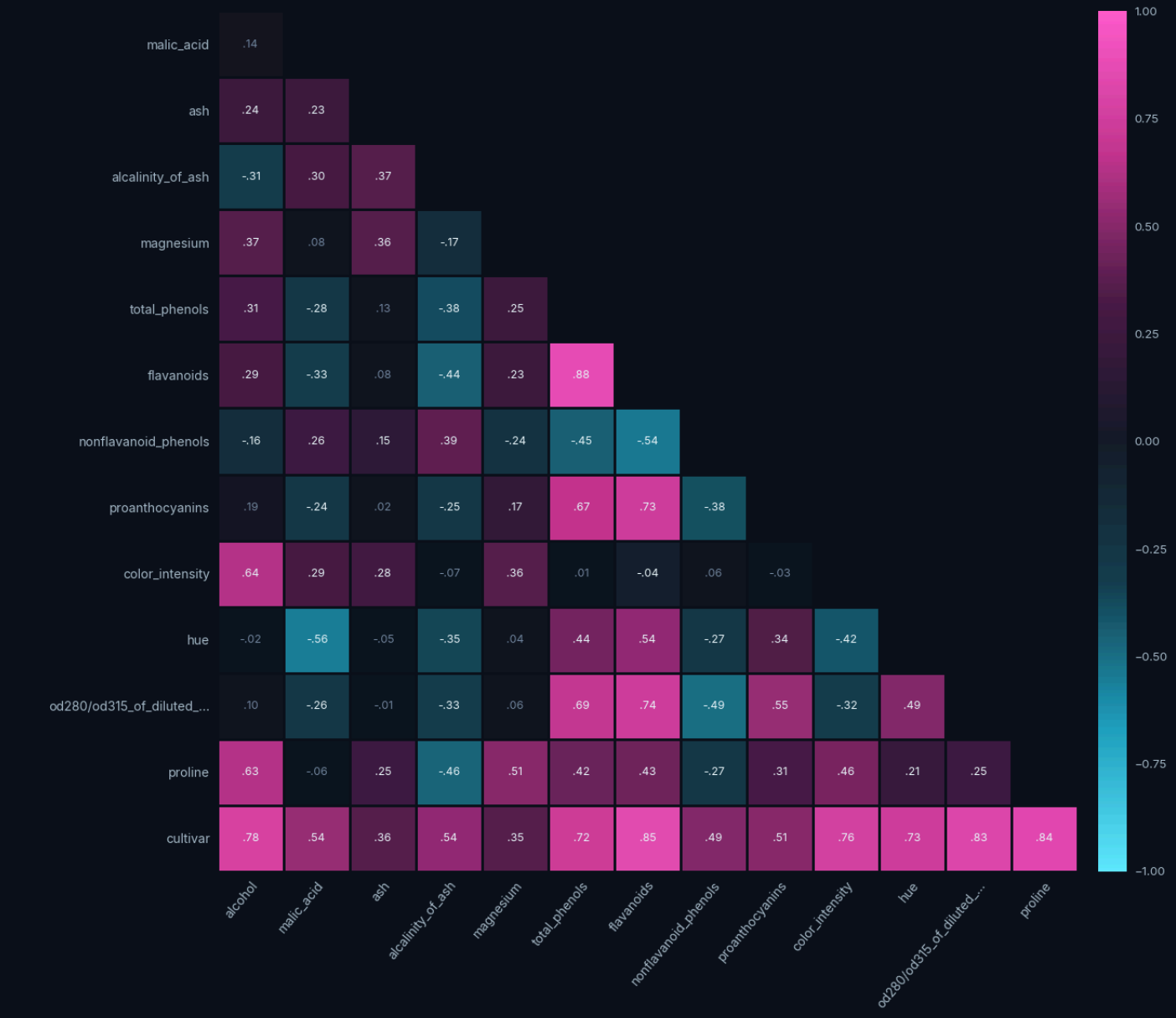
Faded cells: not significant after Benjamini-Hochberg correction (α = 0.05).

All associations

FIGURE // 03_dataset_association.png

All associations

14 columns · Spearman for numbers, Cramér's V for labels, eta across · 74 significant



Faded cells: not significant after Benjamini-Hochberg correction (α = 0.05).

Strongest associations

COLUMN A	COLUMN B	METHOD	VALUE	95% CI	OTHER	N	P_ADJ	SIG
total_phenols	flavanoids	pearson	+0.865	[+0.822, +0.898]	spearman +0.879	178	2e-52	*
cultivar	flavanoids	eta	+0.853			178	1e-27	*
cultivar	proline	eta	+0.839			178	3e-25	*
cultivar	od280/od315_of_diluted_wines	eta	+0.827			178	6e-23	*
flavanoids	od280/od315_of_diluted_wines	pearson	+0.787	[+0.724, +0.837]	spearman +0.742	178	4e-37	*

COLUMN A	COLUMN B	METHOD	VALUE	95% CI	OTHER	N	P_ADJ	SIG
cultivar	alcohol	eta	+0.779			178	2e-23	*
cultivar	color_intensity	eta	+0.761			178	2e-24	*
cultivar	hue	eta	+0.733			178	6e-20	*
cultivar	total_phenols	eta	+0.719			178	4e-20	*
total_phenols	od280/od315_of_diluted_wines	pearson	+0.700	[+0.616, +0.768]	spearman +0.687	178	4e-26	*
flavanoids	proanthocyanins	pearson	+0.653	[+0.559, +0.730]	spearman +0.730	178	6e-22	*
alcohol	proline	pearson	+0.644	[+0.549, +0.722]	spearman +0.634	178	3e-21	*
total_phenols	proanthocyanins	pearson	+0.612	[+0.511, +0.697]	spearman +0.667	178	7e-19	*
hue	od280/od315_of_diluted_wines	pearson	+0.565	[+0.456, +0.658]	spearman +0.485	178	1e-15	*
malic_acid	hue	spearman	-0.560	[-0.660, -0.441]	pearson -0.561	178	3e-15	*
alcohol	color_intensity	pearson	+0.546	[+0.434, +0.642]	spearman +0.635	178	2e-14	*
cultivar	malic_acid	eta	+0.545			178	5e-11	*
flavanoids	hue	pearson	+0.543	[+0.431, +0.639]	spearman +0.535	178	2e-14	*
cultivar	alcalinity_of_ash	eta	+0.539			178	4e-13	*
flavanoids	nonflavanoid_phenols	pearson	-0.538	[-0.635, -0.424]	spearman -0.544	178	5e-14	*
color_intensity	hue	pearson	-0.522	[-0.621, -0.406]	spearman -0.419	178	4e-13	*
proanthocyanins	od280/od315_of_diluted_wines	pearson	+0.519	[+0.403, +0.619]	spearman +0.554	178	5e-13	*
magnesium	proline	spearman	+0.508	[+0.382, +0.615]	pearson +0.393	178	2e-12	*
cultivar	proanthocyanins	eta	+0.507			178	5e-13	*
nonflavanoid_phenols	od280/od315_of_diluted_wines	pearson	-0.503	[-0.606, -0.385]	spearman -0.495	178	3e-12	*

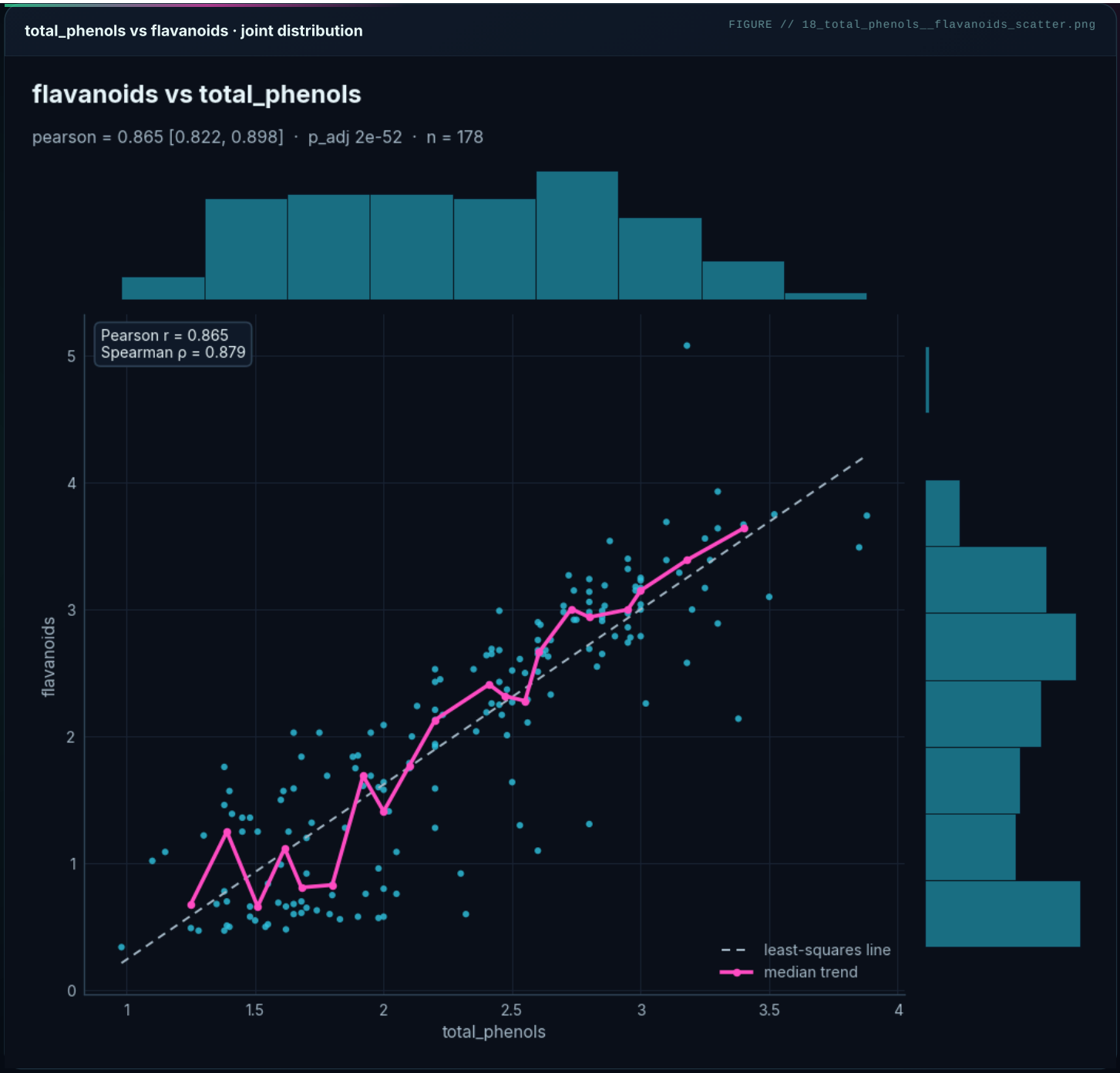
Multicollinearity (VIF)

COLUMN	VIF
`flavanoids`	7.03

COLUMN	VIF
total_phenols	4.33
od280/od315_of_diluted_wines	3.79
color_intensity	3.03
proline	2.82
hue	2.55
alcohol	2.46
alcalinity_of_ash	2.24
ash	2.19
proanthocyanins	1.98
nonflavanoid_phenols	1.80
malic_acid	1.66
magnesium	1.42

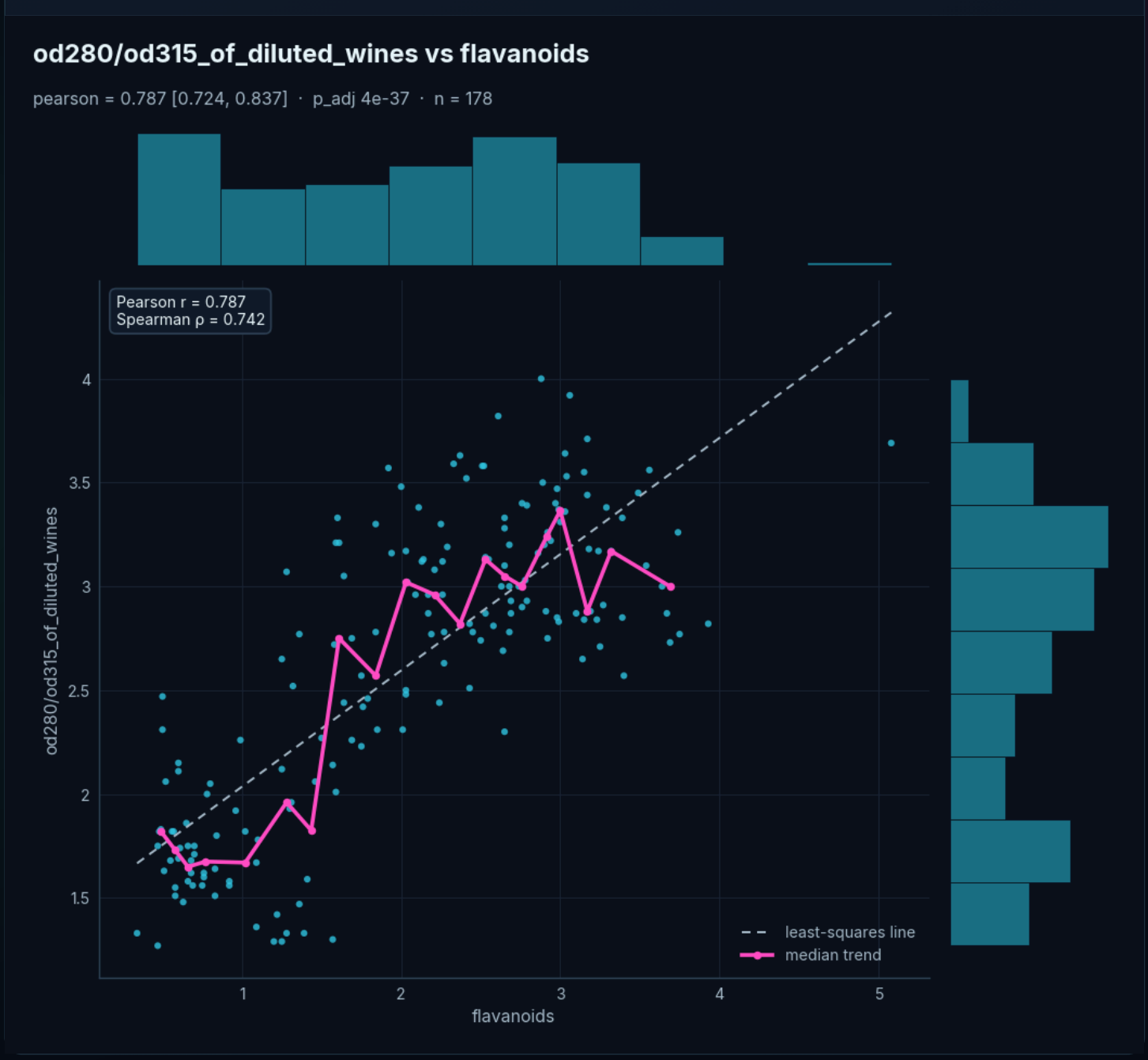
total_phenols vs flavanoids · joint distribution

FIGURE // 18_total_phenols__flavanoids_scatter.png



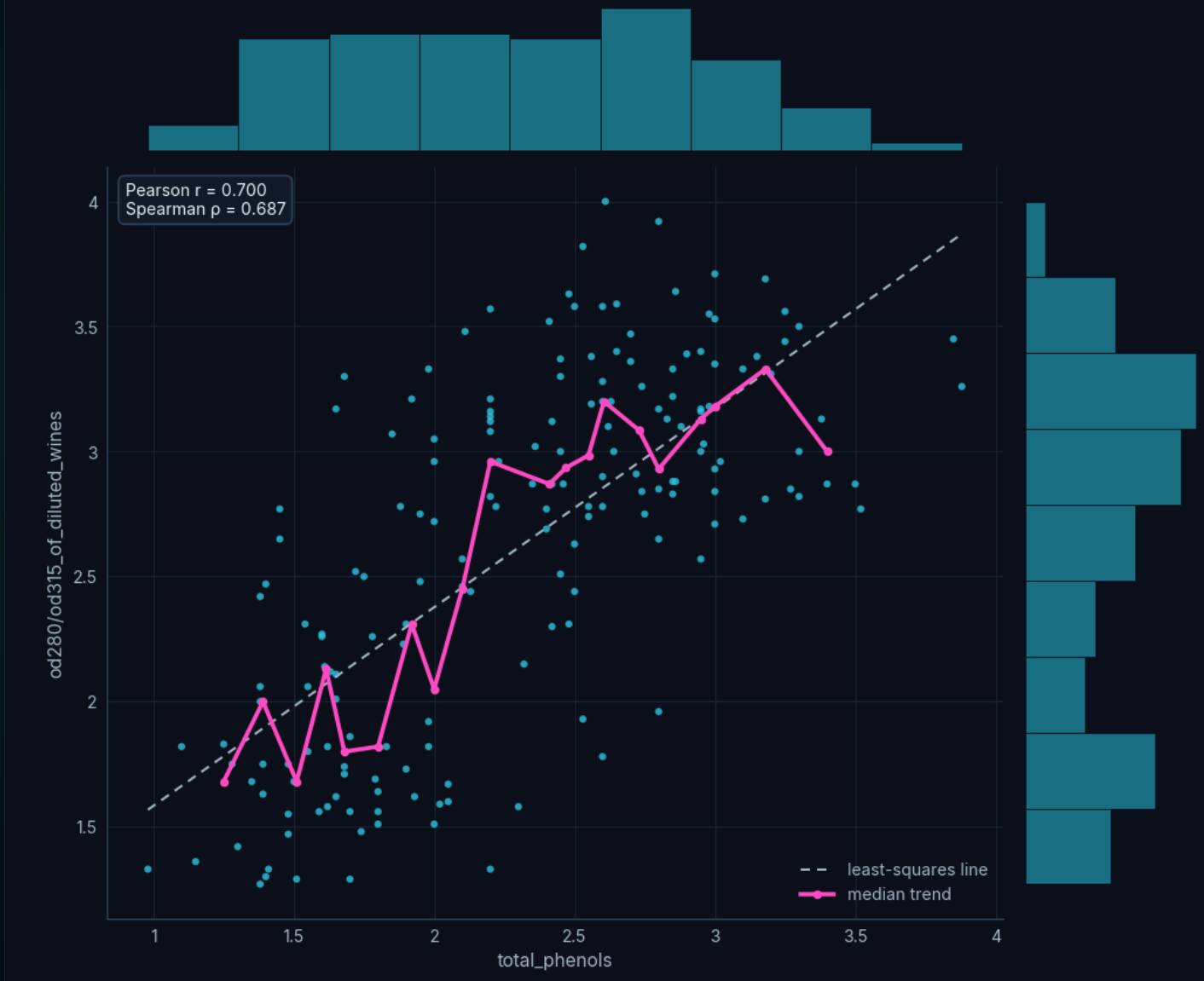
flavanoids vs od280/od315_of_diluted_wines · joint distribution

FIGURE // 19_flavanoids__od280_od315_of_diluted_wines_scatter.png



total_phenols vs od280/od315_of_diluted_wines · joint distribution

FIGURE // 20_total_phenols__od280_od315_of_diluted_wines_scatter…



09 / PREPARATION PLAN

Preparation plan.

Cleaning: reassign outliers 7. Transformation: bucketing 12, combination 1, transformation 2, rescale 1, dimensionality reduction 2.

Cleaning

ACTION	COLUMN	WHY	CODE
Reassign outliers	`malic_acid`	3 consensus outliers form a real but influential tail: cap them at the boxplot fences (winsorize) or use robust models	`df['malic_acid'] = df['malic_acid'].clip(-0.6175, 5.3025)`
Reassign outliers	`ash`	3 consensus outliers form a real but influential tail: cap them at the boxplot fences (winsorize) or use robust models	`df['ash'] = df['ash'].clip(1.68875, 3.07875)`
Reassign outliers	`alcalinity_of_ash`	1 consensus outliers form a real but influential tail: cap them at the boxplot fences (winsorize) or use robust models	`df['alcalinity_of_ash'] = df['alcalinity_of_ash'].clip(10.75, 27.95)`

ACTION	COLUMN	WHY	CODE
Reassign outliers	`magnesium`	2 consensus outliers form a real but influential tail: cap them at the boxplot fences (winsorize) or use robust models	`df['magnesium'] = df['magnesium'].clip(59.5, 135.5)`
Reassign outliers	`proanthocyanins`	1 consensus outliers form a real but influential tail: cap them at the boxplot fences (winsorize) or use robust models	`df['proanthocyanins'] = df['proanthocyanins'].clip(0.2, 3)`
Reassign outliers	`color_intensity`	1 consensus outliers form a real but influential tail: cap them at the boxplot fences (winsorize) or use robust models	`df['color_intensity'] = df['color_intensity'].clip(-1.25, 10.67)`
Reassign outliers	`hue`	1 consensus outliers form a real but influential tail: cap them at the boxplot fences (winsorize) or use robust models	`df['hue'] = df['hue'].clip(0.27625, 1.62625)`

Transformation

ACTION	COLUMN	WHY	CODE
Bucketing	`alcohol`	irregular shape (modes, heaping or heavy skew): 5 bins chosen by a decision tree on 'cultivar'. Learnt from the target on every row: for a model, learn it again on the training rows only, or the validation score is optimistic	`# learnt from the target on all rows: for a model, learn it on the training rows only df['alcohol' + '_bin'] = pd.cut(df['alcohol'], [-inf, 12.185, 12.78, 13.54, 13.89, inf])`
Bucketing	`malic_acid`	irregular shape (modes, heaping or heavy skew): 5 bins chosen by a decision tree on 'cultivar'. Learnt from the target on every row: for a model, learn it again on the training rows only, or the validation score is optimistic	`# learnt from the target on all rows: for a model, learn it on the training rows only df['malic_acid' + '_bin'] = pd.cut(df['malic_acid'], [-inf, 1.475, 1.675, 1.91, 2.455, inf])`
Bucketing	`ash`	irregular shape (modes, heaping or heavy skew): 5 bins chosen by a decision tree on 'cultivar'. Learnt from the target on every row: for a model, learn it again on the training rows only, or the validation score is optimistic	`# learnt from the target on all rows: for a model, learn it on the training rows only df['ash' + '_bin'] = pd.cut(df['ash'], [-inf, 2.03, 2.315, 2.385, 2.535, inf])`
Bucketing	`alcalinity_of_ash`	irregular shape (modes, heaping or heavy skew): 5 bins chosen by a decision tree on 'cultivar'. Learnt from the target on every row: for a model, learn it again on the training rows only, or the validation score is optimistic	`# learnt from the target on all rows: for a model, learn it on the training rows only df['alcalinity_of_ash' + '_bin'] = pd.cut(df['alcalinity_of_ash'], [-inf, 17.9, 19.45, 20.6, 22.9, inf])`
Bucketing	`magnesium`	irregular shape (modes, heaping or heavy skew): 5 bins chosen by a decision tree on 'cultivar'. Learnt from the target on every row: for a model, learn it again on the training rows only, or the validation score is optimistic	`# learnt from the target on all rows: for a model, learn it on the training rows only df['magnesium' + '_bin'] = pd.cut(df['magnesium'], [-inf, 88.5, 99.5, 103.5, 117.5, inf])`
Bucketing	`total_phenols`	irregular shape (modes, heaping or heavy skew): 5 bins chosen by a decision tree on 'cultivar'. Learnt from the target on every row: for a model, learn it again on the training rows only, or the validation score is optimistic	`# learnt from the target on all rows: for a model, learn it on the training rows only df['total_phenols' + '_bin'] = pd.cut(df['total_phenols'], [-inf, 1.595, 1.84, 2.335, 2.58, inf])`

ACTION	COLUMN	WHY	CODE
Bucketing	`flavanoids`	irregular shape (modes, heaping or heavy skew): 5 bins chosen by a decision tree on 'cultivar'. Learnt from the target on every row: for a model, learn it again on the training rows only, or the validation score is optimistic	`# learnt from the target on all rows: for a model, learn it on the training rows only df['flavanoids' + '_bin'] = pd.cut(df['flavanoids'], [-inf, 0.88, 1.4, 2.31, 2.665, inf])`
Bucketing	`nonflavanoid_phenols`	irregular shape (modes, heaping or heavy skew): 5 bins chosen by a decision tree on 'cultivar'. Learnt from the target on every row: for a model, learn it again on the training rows only, or the validation score is optimistic	`# learnt from the target on all rows: for a model, learn it on the training rows only df['nonflavanoid_phenols' + '_bin'] = pd.cut(df['nonflavanoid_phenols'], [-inf, 0.275, 0.335, 0.395, 0.49, inf])`
Bucketing	`proanthocyanins`	irregular shape (modes, heaping or heavy skew): 5 bins chosen by a decision tree on 'cultivar'. Learnt from the target on every row: for a model, learn it again on the training rows only, or the validation score is optimistic	`# learnt from the target on all rows: for a model, learn it on the training rows only df['proanthocyanins' + '_bin'] = pd.cut(df['proanthocyanins'], [-inf, 1.305, 1.655, 1.865, 2.02, inf])`
Bucketing	`hue`	irregular shape (modes, heaping or heavy skew): 5 bins chosen by a decision tree on 'cultivar'. Learnt from the target on every row: for a model, learn it again on the training rows only, or the validation score is optimistic	`# learnt from the target on all rows: for a model, learn it on the training rows only df['hue' + '_bin'] = pd.cut(df['hue'], [-inf, 0.785, 0.925, 1.005, 1.14, inf])`
Bucketing	`od280/od315_of_diluted_wines`	irregular shape (modes, heaping or heavy skew): 5 bins chosen by a decision tree on 'cultivar'. Learnt from the target on every row: for a model, learn it again on the training rows only, or the validation score is optimistic	`# learnt from the target on all rows: for a model, learn it on the training rows only df['od280/od315_of_diluted_wines' + '_bin'] = pd.cut(df['od280/od315_of_diluted_wines'], [-inf, 2.115, 2.545, 3.04, 3.305, inf])`
Bucketing	`proline`	irregular shape (modes, heaping or heavy skew): 5 bins chosen by a decision tree on 'cultivar'. Learnt from the target on every row: for a model, learn it again on the training rows only, or the validation score is optimistic	`# learnt from the target on all rows: for a model, learn it on the training rows only df['proline' + '_bin'] = pd.cut(df['proline'], [-inf, 468.0, 525.0, 755.0, 987.5, inf])`
Combination	`total_phenols, flavanoids, od280/od315_of_diluted_wines`	3 columns move together (mean |Spearman| 0.77): keep 'flavanoids', or combine them into one score	`cols = ['total_phenols', 'flavanoids', 'od280/od315_of_diluted_wines'] df['combined'] = ((df[cols] - df[cols].mean()) / df[cols].std()).mean(axis=1) # or: df = df.drop(columns=['total_phenols', 'od280/od315_of_diluted_wines'])`
Transformation	`malic_acid`	skewness 1.04 → 0.03 after Box-Cox (lambda -0.28)	`from scipy.stats import boxcox _, lam = boxcox(df['malic_acid'].dropna()) # fit (on training rows only for a model) df['malic_acid'] = boxcox(df['malic_acid'], lmbda=lam) # missing values stay missing`

ACTION	COLUMN	WHY	CODE

Transformation	`magnesium`	skewness 1.10 → 0.01 after Box-Cox (lambda -1.43)	`from scipy.stats import boxcox`
			`_, lam = boxcox(df['magnesium'].dropna())  # fit (on training rows only for a model)`
			`df['magnesium'] = boxcox(df['magnesium'], lmbda=lam)  # missing values stay missing`

Rescale	-	numeric scales differ by x2,893 (proline vs nonflavanoid_phenols): distance-based models, regularised models and neural nets need common scales	`from sklearn.preprocessing import RobustScaler`
			`df[['malic_acid', 'ash', 'alcalinity_of_ash', 'magnesium', 'proanthocyanins', 'color_intensity', 'hue']] = RobustScaler().fit_transform(df[['malic_acid', 'ash', 'alcalinity_of_ash', 'magnesium', 'proanthocyanins', 'color_intensity', 'hue']])  # outliers: median and IQR`
			`from sklearn.preprocessing import StandardScaler`
			`df[['alcohol', 'total_phenols', 'flavanoids', 'nonflavanoid_phenols', 'od280/od315_of_diluted_wines', 'proline']] = StandardScaler().fit_transform(df[['alcohol', 'total_phenols', 'flavanoids', 'nonflavanoid_phenols', 'od280/od315_of_diluted_wines', 'proline']])`

Dimensionality reduction	-	PCA: 3 components (parallel analysis) keep 67% of the variance of 13 columns (KMO 0.78 middling, Bartlett p 2e-224)	`from sklearn.pipeline import make_pipeline`
			`from sklearn.impute import SimpleImputer`
			`from sklearn.preprocessing import StandardScaler`
			`from sklearn.decomposition import PCA`
			`cols = ['alcohol', 'malic_acid', 'ash', 'alcalinity_of_ash', 'magnesium', 'total_phenols', 'flavanoids', 'nonflavanoid_phenols', 'proanthocyanins', 'color_intensity', 'hue', 'od280/od315_of_diluted_wines', 'proline']`
			`pca = make_pipeline(SimpleImputer(strategy='median'), StandardScaler(), PCA(n_components=3))`
			`scores = pca.fit_transform(df[cols])`

ACTION	COLUMN	WHY	CODE

Dimensionality reduction	-	Factor analysis: 3 factors explain 70% of the variance of 12 variables (KMO 0.79, Bartlett p 4e-220); 3 reliable scales created	`items = df[['flavanoids', 'od280/od315_of_diluted_wines', 'total_phenols', 'hue', 'proanthocyanins', 'nonflavanoid_phenols', 'malic_acid']].copy()`
			`items = (items - items.mean()) / items.std()  # different scales: z-scores`
			`items['nonflavanoid_phenols'] = -items['nonflavanoid_phenols']  # reversed item (negative loading)`
			`items['malic_acid'] = -items['malic_acid']  # reversed item (negative loading)`
			`df['F1_flavanoids_od280/od315_of_diluted_wines'] = items.mean(axis=1)  # MEAN: the items answered`
			`# MEAN.7 (every item required): items.mean(axis=1).where(items.notna().all(axis=1))`
			
			`items = df[['alcohol', 'color_intensity', 'proline']].copy()`
			`items = (items - items.mean()) / items.std()  # different scales: z-scores`
			`df['F2_alcohol_color_intensity'] = items.mean(axis=1)  # MEAN: the items answered`
			`# MEAN.3 (every item required): items.mean(axis=1).where(items.notna().all(axis=1))`
			
			`items = df[['ash', 'alcalinity_of_ash']].copy()`
			`items = (items - items.mean()) / items.std()  # different scales: z-scores`
			`df['F3_ash_alcalinity_of_ash'] = items.mean(axis=1)  # MEAN: the items answered`
			`# MEAN.2 (every item required): items.mean(axis=1).where(items.notna().all(axis=1))`

10 / ATTRIBUTE STATISTICS

Attribute statistics.

COLUMN	TYPE	MISSING	DISTINCT	ALERTS
`alcohol`	numeric_continuous	0.0%	126	0
`malic_acid`	numeric_continuous	0.0%	133	0
`ash`	numeric_continuous	0.0%	79	0
`alcalinity_of_ash`	numeric_continuous	0.0%	63	1
`magnesium`	numeric_continuous	0.0%	53	0
`total_phenols`	numeric_continuous	0.0%	97	1
`flavanoids`	numeric_continuous	0.0%	132	0

COLUMN	TYPE	MISSING	DISTINCT	ALERTS
nonflavanoid_phenols	numeric_continuous	0.0%	39	0
proanthocyanins	numeric_continuous	0.0%	101	0
color_intensity	numeric_continuous	0.0%	132	0
hue	numeric_continuous	0.0%	78	0
od280/od315_of_diluted_wines	numeric_continuous	0.0%	122	0
proline	numeric_continuous	0.0%	121	1
cultivar	categorical	0.0%	3	0

alcohol · numeric_continuous

float64 · float64 dtype: 126 distinct real values

Decision path: any non-missing value? → a single distinct value? → how is it stored? (float64) → only 0 and 1? → dates written as numbers (YYYYMMDD, Unix time…)? → an identifier by its name? → unique, consecutive or sorted integers? → integers that label categories (codes)? → a small scale of consecutive integers? → integers? → a handful of distinct real values? → **real numbers yes: numeric_continuous** (certain: decided on every value)

PROFILE	
type	numeric_continuous
dtype	float64
values	178
missing	0 (0.0%)
distinct	126 (70.8%)
memory	1.4 KB
mode	13.05 ×6

CENTER	
mean	13
95% CI of mean	[12.88, 13.12]
median	13.05
95% CI of median	[12.81, 13.23]
trimmed mean	13.01
geometric mean	12.98
harmonic mean	12.95

SPREAD	
std	0.8118
variance	0.6591
sem	0.06085
IQR	1.315
MAD	0.68
CV	0.06245
range	3.8

QUANTILES	
min	11.03
p1	11.44
p5	11.66
p25	12.36
p50	13.05
p75	13.68
p95	14.22
p99	14.47
max	14.83
sum	2,314.1

SHAPE	
skewness	-0.0515
excess kurtosis	-0.852
KDE modes	1
bimodality coef.	0.456
decimals	2
heaping	×1.0

OUTLIERS & COUNTS	
IQR fences	[10.39, 15.65]
outside fences	0 (0.0%)
|z| > 3	0 (0.0%)
modified z > 3.5	0 (0.0%)
zeros	0
negatives	0
infinite	0

ROW ORDER	
monotonic	no
lag-1 autocorrelation	0.650 (p 4e-18)

Normality: not normal (n = 178)

TEST	STATISTIC	P-VALUE	AT ALPHA 0.05
Shapiro-Wilk	0.9818	0.02	rejects normality
D'Agostino K^2	15.96	3e-04	rejects normality
Jarque-Bera	5.592	0.061	consistent with normal
Anderson-Darling	1.034	0.01	rejects normality

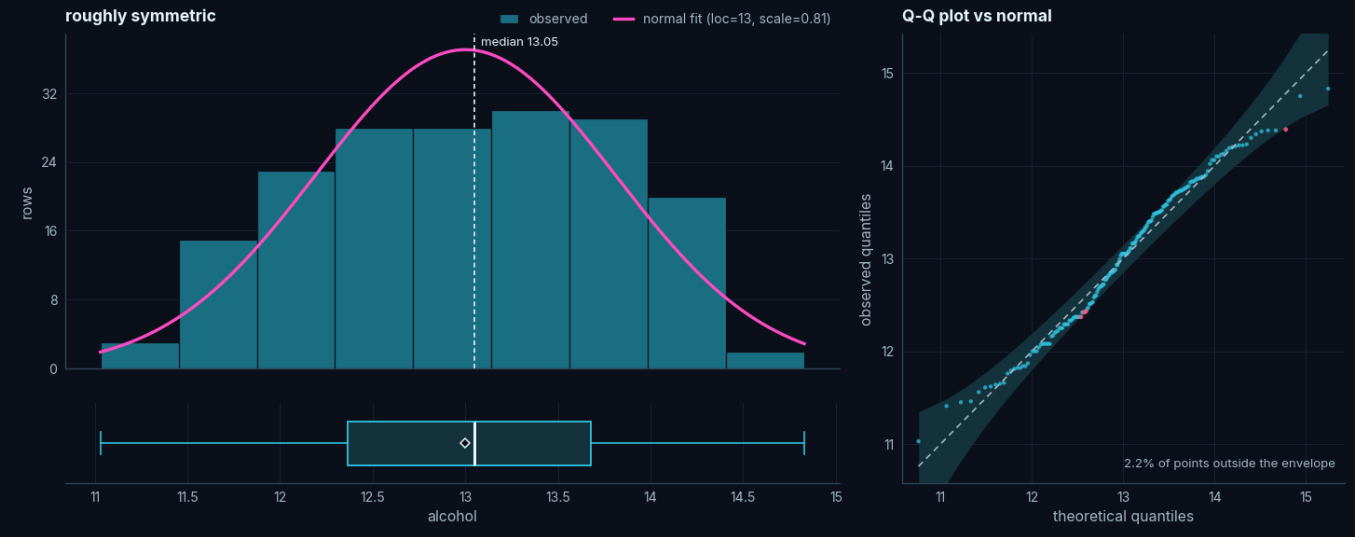
Best fits (continuous, 13 candidates fitted on 178 values)

#	DISTRIBUTION	PARAMETERS	AIC	ΔAIC	WEIGHT	BIC	KS D	KS P
1	normal	loc=13, scale=0.81	433.924	0.0	0.28	440.288	0.069 (fair)	0.36
2	chi-squared	df=513, scale=0.0253	434.468	0.5	0.21	440.831	0.070 (fair)	0.34
3	gamma	a=256, scale=0.0507	434.468	0.5	0.21	440.831	0.070 (fair)	0.34

alcohol · distribution FIGURE // 04_alcohol_dist.png

alcohol

numeric_continuous · n = 178 · mean 13 · median 13.05 · sd 0.8118 · skew -0.05 · not normal



Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

`malic_acid` **· numeric_continuous** −

float64 · float64 dtype: 133 distinct real values

Decision path: any non-missing value? → a single distinct value? → how is it stored? (float64) → only 0 and 1? → dates written as numbers (YYYYMMDD, Unix time…)? → an identifier by its name? → unique, consecutive or sorted integers? → integers that label categories (codes)? → a small scale of consecutive integers? → integers? → a handful of distinct real values? → **real numbers yes: numeric_continuous** (certain: decided on every value)

PROFILE	
type	numeric_continuous
dtype	float64
values	178
missing	0 (0.0%)
distinct	133 (74.7%)
memory	1.4 KB
mode	1.73 ×7

CENTER	
mean	2.336
95% CI of mean	[2.171, 2.502]
median	1.865
95% CI of median	[1.73, 2.05]
trimmed mean	2.212
geometric mean	2.107
harmonic mean	1.914

SPREAD	
std	1.117
variance	1.248
sem	0.08373
IQR	1.48
MAD	0.52
CV	0.4782
range	5.06

QUANTILES	
min	0.74
p1	0.8977
p5	1.061
p25	1.603
p50	1.865
p75	3.083
p95	4.456
p99	5.542
max	5.8
sum	415.9

SHAPE	
skewness	1.04
excess kurtosis	0.299
KDE modes	1
bimodality coef.	0.621
decimals	2
heaping	×1.1

OUTLIERS & COUNTS	
IQR fences	[-0.6175, 5.303]
outside fences	3 (1.7%)
|z| > 3	1 (0.6%)
modified z > 3.5	9 (5.1%)
zeros	0
negatives	0
infinite	0

ROW ORDER	
monotonic	no
lag-1 autocorrelation	0.420 (p 2e-08)

Normality: not normal (n = 178)

TEST	STATISTIC	P-VALUE	AT ALPHA 0.05
Shapiro-Wilk	0.8888	3e-10	rejects normality
D'Agostino K^2	25.32	3e-06	rejects normality
Jarque-Bera	32.02	1e-07	rejects normality
Anderson-Darling	7.619	0.01	rejects normality

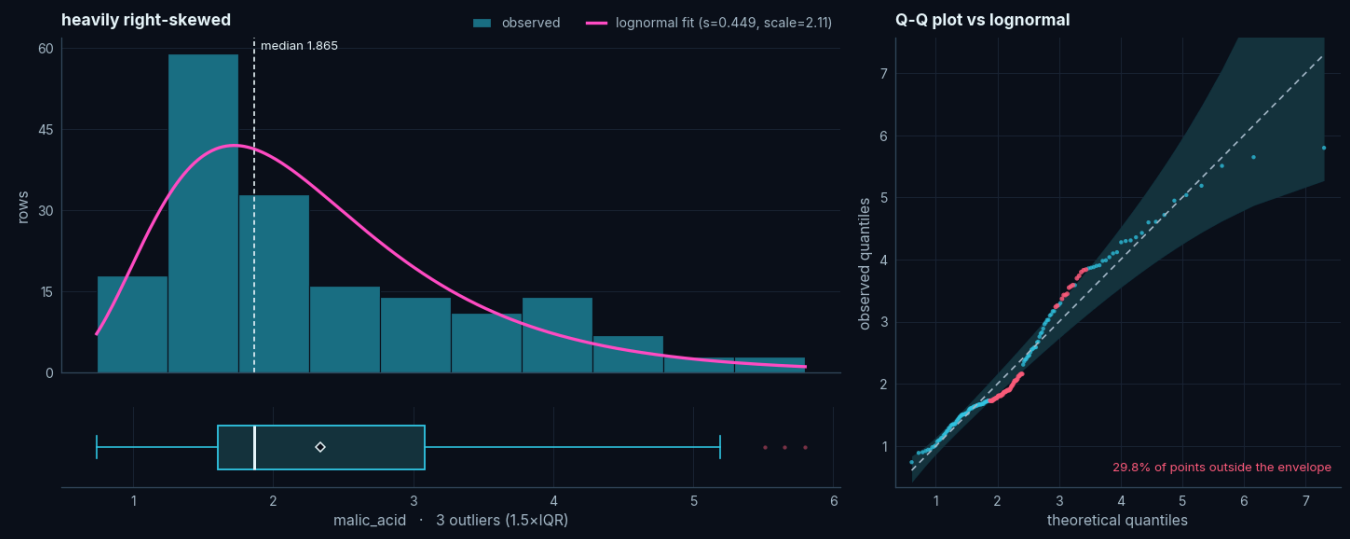
Best fits (continuous, 13 candidates fitted on 178 values)

#	DISTRIBUTION	PARAMETERS	AIC	ΔAIC	WEIGHT	BIC	KS D	KS P
1	lognormal	s=0.449, scale=2.11	489.326	0.0	0.99	495.689	0.130 (poor)	0.0043
2	gamma	a=4.99, scale=0.468	500.122	10.8	0.00	506.486	0.155 (poor)	3e-04
3	chi-squared	df=9.98, scale=0.234	500.122	10.8	0.00	506.486	0.155 (poor)	3e-04

malic_acid · distribution FIGURE // 05_malic_acid_dist.png

malic_acid

numeric_continuous · n = 178 · mean 2.336 · median 1.865 · sd 1.117 · skew 1.04 · not normal



Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

`ash` **· numeric_continuous**

float64 · float64 dtype: 79 distinct real values

Decision path: any non-missing value? → a single distinct value? → how is it stored? (float64) → only 0 and 1? → dates written as numbers (YYYYMMDD, Unix time…)? → an identifier by its name? → unique, consecutive or sorted integers? → integers that label categories (codes)? → a small scale of consecutive integers? → integers? → a handful of distinct real values? → **real numbers yes: numeric_continuous** (certain: decided on every value)

PROFILE	
type	numeric_continuous
dtype	float64
values	178
missing	0 (0.0%)
distinct	79 (44.4%)
memory	1.4 KB
mode	2.3 ×7

CENTER	
mean	2.367
95% CI of mean	[2.326, 2.407]
median	2.36
95% CI of median	[2.3, 2.4]
trimmed mean	2.374
geometric mean	2.35
harmonic mean	2.333

SPREAD	
std	0.2743
variance	0.07526
sem	0.02056
IQR	0.3475
MAD	0.16
CV	0.1159
range	1.87

QUANTILES	
min	1.36
p1	1.7
p5	1.92
p25	2.21
p50	2.36
p75	2.558
p95	2.742
p99	2.989
max	3.23
sum	421.2

SHAPE	
skewness	-0.177
excess kurtosis	1.14
KDE modes	1
bimodality coef.	0.246
decimals	2
heaping	×1.5

OUTLIERS & COUNTS	
IQR fences	[1.689, 3.079]
outside fences	3 (1.7%)
|z| > 3	3 (1.7%)
modified z > 3.5	3 (1.7%)
zeros	0
negatives	0
infinite	0

ROW ORDER	
monotonic	no
lag-1 autocorrelation	0.200 (p 0.0077)

Normality: not normal (n = 178)

TEST	STATISTIC	P-VALUE	AT ALPHA 0.05
Shapiro-Wilk	0.9839	0.039	rejects normality
D'Agostino K^2	6.631	0.036	rejects normality
Jarque-Bera	9.539	0.0085	rejects normality
Anderson-Darling	0.6784	0.079	consistent with normal

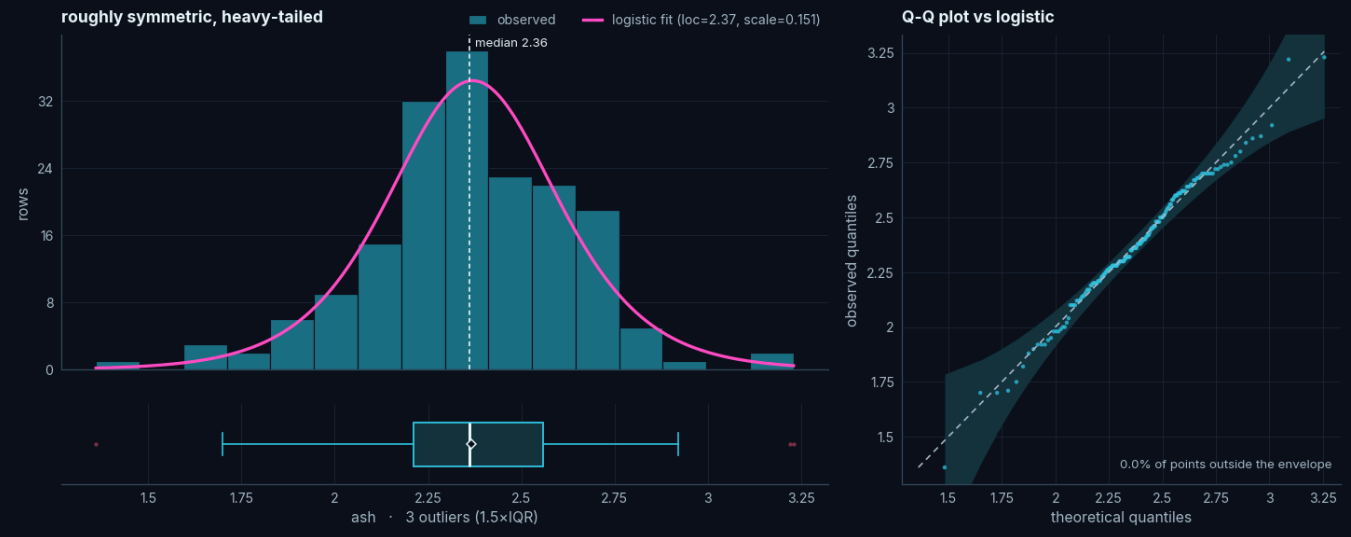
Best fits (continuous, 13 candidates fitted on 178 values)

#	DISTRIBUTION	PARAMETERS	AIC	ΔAIC	WEIGHT	BIC	KS D	KS P
1	logistic	loc=2.37, scale=0.151	42.8361	0.0	0.66	49.1997	0.048 (fair)	0.79
2	Student t	df=7.93, loc=2.37, scale=0.237	44.8248	2.0	0.25	54.3701	0.047 (fair)	0.8
3	normal	loc=2.37, scale=0.274	47.6987	4.9	0.06	54.0623	0.058 (fair)	0.57

ash · distribution FIGURE // 06_ash_dist.png

ash

numeric_continuous · n = 178 · mean 2.367 · median 2.36 · sd 0.2743 · skew -0.18 · not normal



Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

`alcalinity_of_ash` · **numeric_continuous · 1 alert** —

float64 · float64 dtype: 63 distinct real values

Decision path: any non-missing value? → a single distinct value? → how is it stored? (float64) → only 0 and 1? → dates written as numbers (YYYYMMDD, Unix time…)? → an identifier by its name? → unique, consecutive or sorted integers? → integers that label categories (codes)? → a small scale of consecutive integers? → integers? → a handful of distinct real values? → **real numbers yes: numeric_continuous** (certain: decided on every value)

- `INFO` heaping: 75% of values end in 0 or 5 (expected ~20%) → rounded / self-reported values?

PROFILE	
type	numeric_continuous
dtype	float64
values	178
missing	0 (0.0%)
distinct	63 (35.4%)
memory	1.4 KB
mode	20.0 ×15

CENTER	
mean	19.49
95% CI of mean	[19, 19.99]
median	19.5
95% CI of median	[18.8, 20]
trimmed mean	19.42
geometric mean	19.2
harmonic mean	18.9

SPREAD	
std	3.34
variance	11.15
sem	0.2503
IQR	4.3
MAD	2.05
CV	0.1713
range	19.4

QUANTILES	
min	10.6
p1	11.35
p5	14.77
p25	17.2
p50	19.5
p75	21.5
p95	25
p99	28.5
max	30
sum	3,470.1

SHAPE	
skewness	0.213
excess kurtosis	0.488
KDE modes	1
bimodality coef.	0.295
decimals	1
heaping	×3.8

OUTLIERS & COUNTS	
IQR fences	[10.75, 27.95]
outside fences	4 (2.2%)
|z| > 3	1 (0.6%)
modified z > 3.5	0 (0.0%)
zeros	0
negatives	0
infinite	0

ROW ORDER	
monotonic	no
lag-1 autocorrelation	0.495 (p 4e-11)

Normality: normal (n = 178)

TEST	STATISTIC	P-VALUE	AT ALPHA 0.05
Shapiro-Wilk	0.9902	0.26	consistent with normal
D'Agostino K^2	3.121	0.21	consistent with normal
Jarque-Bera	2.765	0.25	consistent with normal
Anderson-Darling	0.5008	0.15	consistent with normal

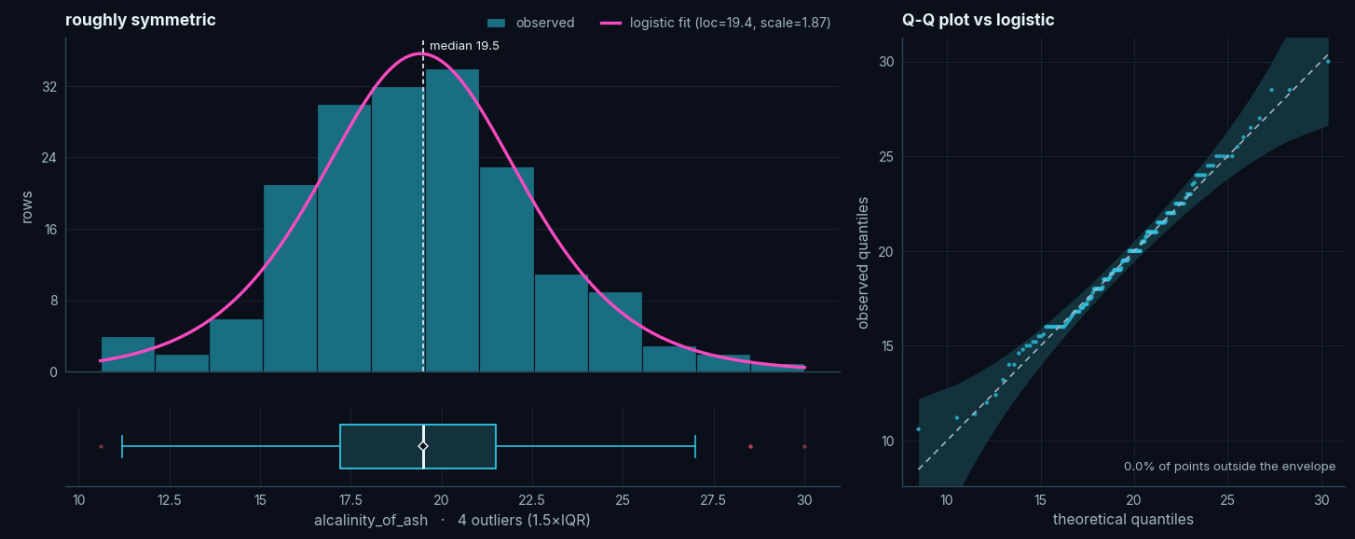
Best fits (continuous, 13 candidates fitted on 178 values)

#	DISTRIBUTION	PARAMETERS	AIC	ΔAIC	WEIGHT	BIC	KS D	KS P
1	logistic	loc=19.4, scale=1.87	936.247	0.0	0.34	942.611	0.047 (fair)	0.81
2	normal	loc=19.5, scale=3.33	937.418	1.2	0.19	943.782	0.063 (fair)	0.45
3	gamma	a=33.5, scale=0.582	937.843	1.6	0.15	944.206	0.052 (fair)	0.71

alcalinity_of_ash · distribution FIGURE // 07_alcalinity_of_ash_dist.png

alcalinity_of_ash

numeric_continuous · n = 178 · mean 19.49 · median 19.5 · sd 3.34 · skew 0.21 · normal



Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

`magnesium` · **numeric_continuous**

float64 · float64 dtype: integer-valued with many levels (53)

Decision path: any non-missing value? → a single distinct value? → how is it stored? (float64) → only 0 and 1? → dates written as numbers (YYYYMMDD, Unix time…)? → an identifier by its name? → unique, consecutive or sorted integers? → integers that label categories (codes)? → a small scale of consecutive integers? → **integers? yes: numeric_continuous** (certain: decided on every value)

PROFILE	
type	numeric_continuous
dtype	float64
values	178
missing	0 (0.0%)
distinct	53 (29.8%)
memory	1.4 KB
mode	88.0 ×13

CENTER	
mean	99.74
95% CI of mean	[97.63, 101.9]
median	98
95% CI of median	[96, 101]
trimmed mean	98.44
geometric mean	98.79
harmonic mean	97.91

SPREAD	
std	14.28
variance	204
sem	1.071
IQR	19
MAD	10
CV	0.1432
range	92

QUANTILES	
min	70
p1	78
p5	80.85
p25	88
p50	98
p75	107
p95	124.3
p99	141.8
max	162
sum	17,754

SHAPE	
skewness	1.1
excess kurtosis	2.1
KDE modes	1
bimodality coef.	0.428
decimals	0
heaping	×0.9

OUTLIERS & COUNTS	
IQR fences	[59.5, 135.5]
outside fences	4 (2.2%)
|z| > 3	2 (1.1%)
modified z > 3.5	2 (1.1%)
zeros	0
negatives	0
infinite	0

ROW ORDER	
monotonic	no
lag-1 autocorrelation	0.367 (p 1e-06)

Normality: not normal (n = 178)

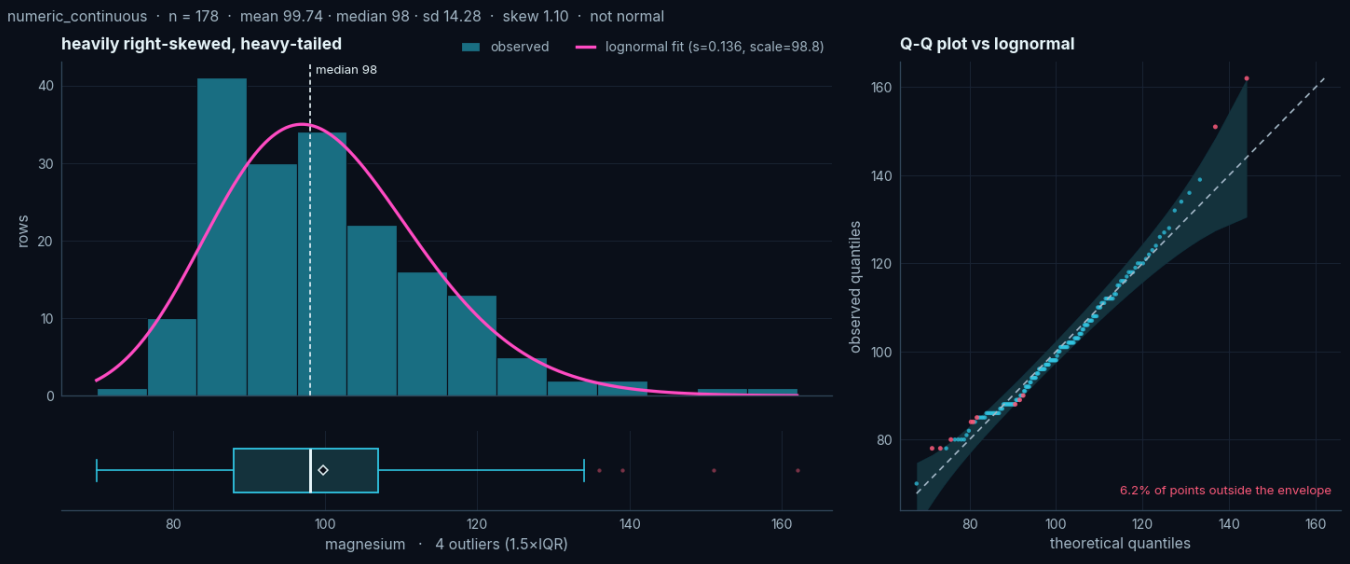
TEST	STATISTIC	P-VALUE	AT ALPHA 0.05
Shapiro-Wilk	0.9383	6e-07	rejects normality
D'Agostino K^2	38.37	5e-09	rejects normality
Jarque-Bera	65.22	7e-15	rejects normality
Anderson-Darling	2.331	0.01	rejects normality

Best fits (continuous, 13 candidates fitted on 178 values)

#	DISTRIBUTION	PARAMETERS	AIC	ΔAIC	WEIGHT	BIC	KS D	KS P
1	lognormal	s=0.136, scale=98.8	1,434.6	0.0	0.88	1,441.0	0.070 (fair)	0.33
2	gamma	a=52.6, scale=1.9	1,440.1	5.5	0.06	1,446.5	0.072 (fair)	0.3
3	chi-squared	df=105, scale=0.949	1,440.1	5.5	0.06	1,446.5	0.072 (fair)	0.3

magnesium · distribution

FIGURE // 08_magnesium_dist.png



Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

`total_phenols` **· numeric_continuous · 1 alert**

float64 · float64 dtype: 97 distinct real values

Decision path: any non-missing value? → a single distinct value? → how is it stored? (float64) → only 0 and 1? → dates written as numbers (YYYYMMDD, Unix time…)? → an identifier by its name? → unique, consecutive or sorted integers? → integers that label categories (codes)? → a small scale of consecutive integers? → integers? → a handful of distinct real values? → **real numbers yes: numeric_continuous** (certain: decided on every value)

- `INFO` heaping: 59% of values end in 0 or 5 (expected ~20%) → rounded / self-reported values?

PROFILE	
type	numeric_continuous
dtype	float64
values	178
missing	0 (0.0%)
distinct	97 (54.5%)
memory	1.4 KB
mode	2.2 ×8

CENTER	
mean	2.295
95% CI of mean	[2.203, 2.388]
median	2.355
95% CI of median	[2.11, 2.48]
trimmed mean	2.286
geometric mean	2.206
harmonic mean	2.114

SPREAD	
std	0.6259
variance	0.3917
sem	0.04691
IQR	1.057
MAD	0.505
CV	0.2727
range	2.9

QUANTILES	
min	0.98
p1	1.138
p5	1.38
p25	1.742
p50	2.355
p75	2.8
p95	3.275
p99	3.596
max	3.88
sum	408.5

SHAPE	
skewness	0.0866
excess kurtosis	-0.836
KDE modes	2
bimodality coef.	0.455
decimals	2
heaping	×2.9

OUTLIERS & COUNTS	
IQR fences	[0.1563, 4.386]
outside fences	0 (0.0%)
$\|z\| > 3$	0 (0.0%)
modified z > 3.5	0 (0.0%)
zeros	0
negatives	0
infinite	0

ROW ORDER	
monotonic	no
lag-1 autocorrelation	0.601 (p 1e-15)

Normality: not normal (n = 178)

TEST	STATISTIC	P-VALUE	AT ALPHA 0.05
Shapiro-Wilk	0.9767	0.0044	rejects normality
D'Agostino K^2	14.99	6e-04	rejects normality
Jarque-Bera	5.525	0.063	consistent with normal
Anderson-Darling	1.421	0.01	rejects normality

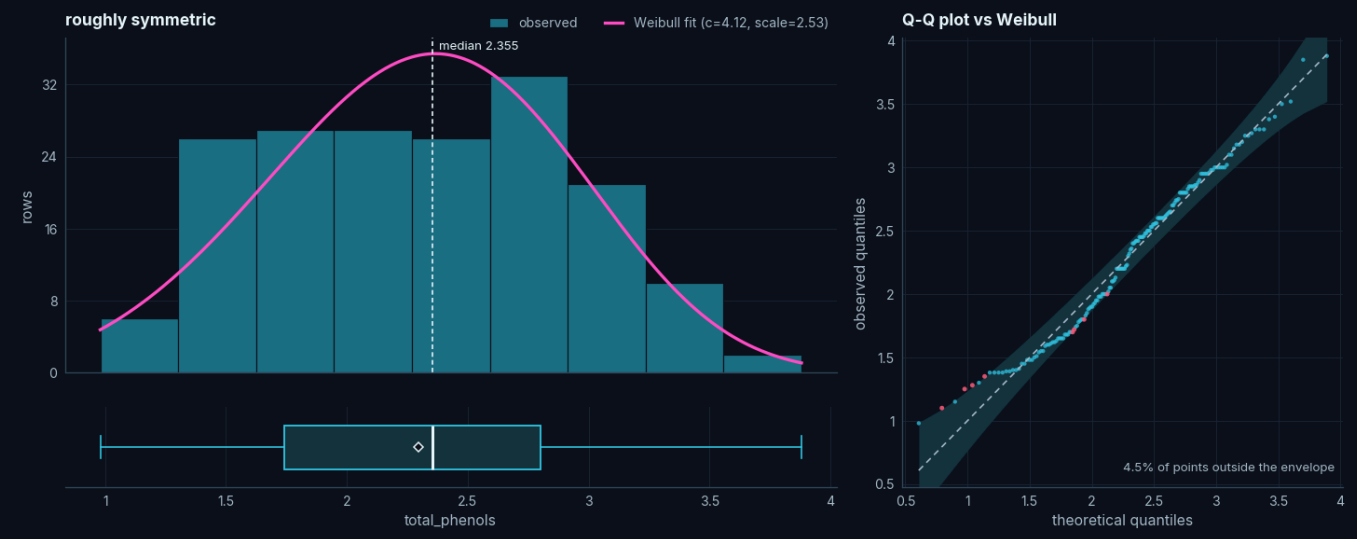
Best fits (continuous, 13 candidates fitted on 178 values)

#	DISTRIBUTION	PARAMETERS	AIC	ΔAIC	WEIGHT	BIC	KS D	KS P
1	Weibull	c=4.12, scale=2.53	339.516	0.0	0.42	345.88	0.072 (fair)	0.3
2	normal	loc=2.3, scale=0.624	341.302	1.8	0.17	347.666	0.073 (fair)	0.29
3	gamma	a=12.8, scale=0.179	341.488	2.0	0.16	347.852	0.095 (poor)	0.077

total_phenols · distribution FIGURE // 09_total_phenols_dist.png

total_phenols

numeric_continuous · n = 178 · mean 2.295 · median 2.355 · sd 0.6259 · skew 0.09 · not normal



Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

`flavanoids` **· numeric_continuous**

float64 · float64 dtype: 132 distinct real values

Decision path: any non-missing value? → a single distinct value? → how is it stored? (float64) → only 0 and 1? → dates written as numbers (YYYYMMDD, Unix time…)? → an identifier by its name? → unique, consecutive or sorted integers? → integers that label categories (codes)? → a small scale of consecutive integers? → integers? → a handful of distinct real values? → **real numbers yes: numeric_continuous** (certain: decided on every value)

PROFILE	
type	numeric_continuous
dtype	float64
values	178
missing	0 (0.0%)
distinct	132 (74.2%)
memory	1.4 KB
mode	2.65 ×4

CENTER	
mean	2.029
95% CI of mean	[1.882, 2.177]
median	2.135
95% CI of median	[1.79, 2.37]
trimmed mean	2.02
geometric mean	1.726
harmonic mean	1.393

SPREAD	
std	0.9989
variance	0.9977
sem	0.07487
IQR	1.67
MAD	0.835
CV	0.4922
range	4.74

QUANTILES	
min	0.34
p1	0.47
p5	0.5455
p25	1.205
p50	2.135
p75	2.875
p95	3.498
p99	3.791
max	5.08
sum	361.2

SHAPE	
skewness	0.0253
excess kurtosis	-0.88
KDE modes	2
bimodality coef.	0.461
decimals	2
heaping	×1.1

OUTLIERS & COUNTS	
IQR fences	[-1.3, 5.38]
outside fences	0 (0.0%)
|z| > 3	1 (0.6%)
modified z > 3.5	0 (0.0%)
zeros	0
negatives	0
infinite	0

ROW ORDER	
monotonic	no
lag-1 autocorrelation	0.788 (p 8e-26)

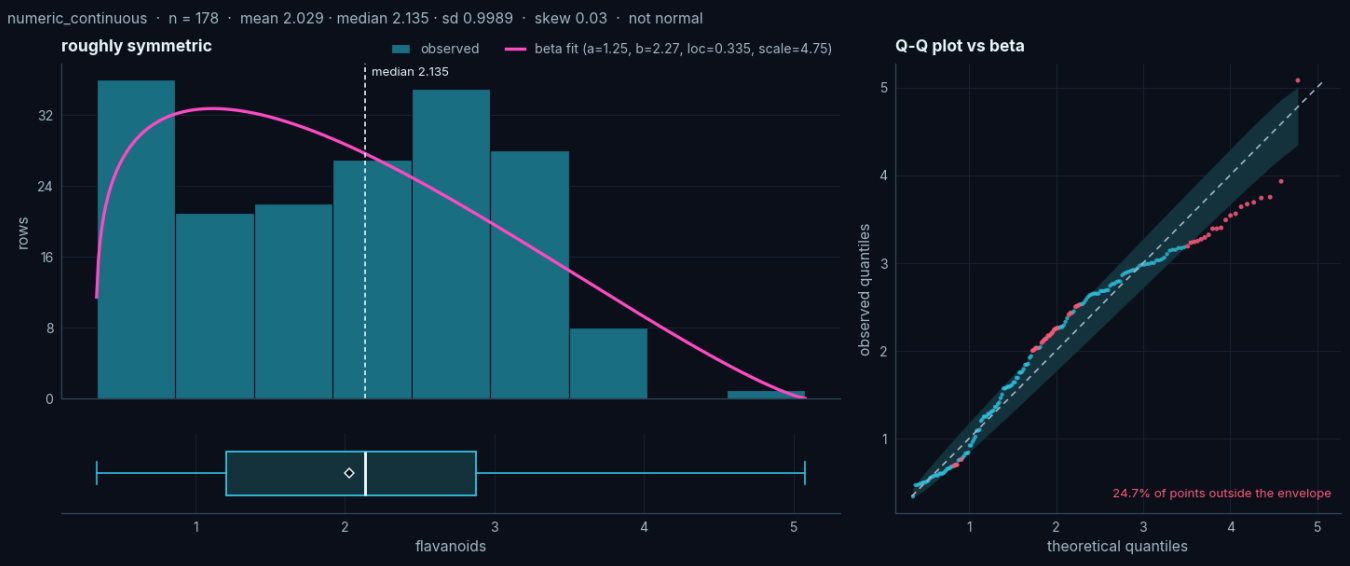
Normality: not normal (n = 178)

TEST	STATISTIC	P-VALUE	AT ALPHA 0.05
Shapiro-Wilk	0.9545	2e-05	rejects normality
D'Agostino K^2	17.94	1e-04	rejects normality
Jarque-Bera	5.885	0.053	consistent with normal
Anderson-Darling	2.474	0.01	rejects normality

Best fits (continuous, 13 candidates fitted on 178 values)

#	DISTRIBUTION	PARAMETERS	AIC	ΔAIC	WEIGHT	BIC	KS D	KS P
1	beta	a=1.25, b=2.27, loc=0.335, scale=4.75	497.575	0.0	0.55	510.302	0.088 (poor)	0.12
2	Weibull	c=2.16, scale=2.29	498.012	0.4	0.44	504.375	0.100 (poor)	0.054
3	normal	loc=2.03, scale=0.996	507.733	10.2	0.00	514.096	0.086 (poor)	0.14

flavanoids · distribution FIGURE // 10_flavanoids_dist.png



Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

`nonflavanoid_phenols` **· numeric_continuous** —

float64 · float64 dtype: 39 distinct real values

Decision path: any non-missing value? → a single distinct value? → how is it stored? (float64) → only 0 and 1? → dates written as numbers (YYYYMMDD, Unix time…)? → an identifier by its name? → unique, consecutive or sorted integers? → integers that label categories (codes)? → a small scale of consecutive integers? → integers? → a handful of distinct real values? → **real numbers yes: numeric_continuous** (certain: decided on every value)

PROFILE	
type	numeric_continuous
dtype	float64
values	178
missing	0 (0.0%)
distinct	39 (21.9%)
memory	1.4 KB
mode	0.26 ×11

CENTER	
mean	0.3619
95% CI of mean	[0.3434, 0.3803]
median	0.34
95% CI of median	[0.3, 0.37]
trimmed mean	0.3551
geometric mean	0.3406
harmonic mean	0.3196

SPREAD	
std	0.1245
variance	0.01549
sem	0.009328
IQR	0.1675
MAD	0.085
CV	0.3439
range	0.53

QUANTILES	
min	0.13
p1	0.14
p5	0.19
p25	0.27
p50	0.34
p75	0.4375
p95	0.6
p99	0.63
max	0.66
sum	64.41

SHAPE	
skewness	0.45
excess kurtosis	-0.637
KDE modes	1
bimodality coef.	0.498
decimals	2
heaping	×0.9

OUTLIERS & COUNTS	
IQR fences	[0.01875, 0.6887]
outside fences	0 (0.0%)
|z| > 3	0 (0.0%)
modified z > 3.5	0 (0.0%)
zeros	0
negatives	0
infinite	0

ROW ORDER	
monotonic	no
lag-1 autocorrelation	0.431 (p 9e-09)

Normality: not normal (n = 178)

TEST	STATISTIC	P-VALUE	AT ALPHA 0.05
Shapiro-Wilk	0.9625	1e-04	rejects normality
D'Agostino K^2	11.86	0.0027	rejects normality
Jarque-Bera	9.072	0.011	rejects normality
Anderson-Darling	2.172	0.01	rejects normality

Best fits (continuous, 13 candidates fitted on 178 values)

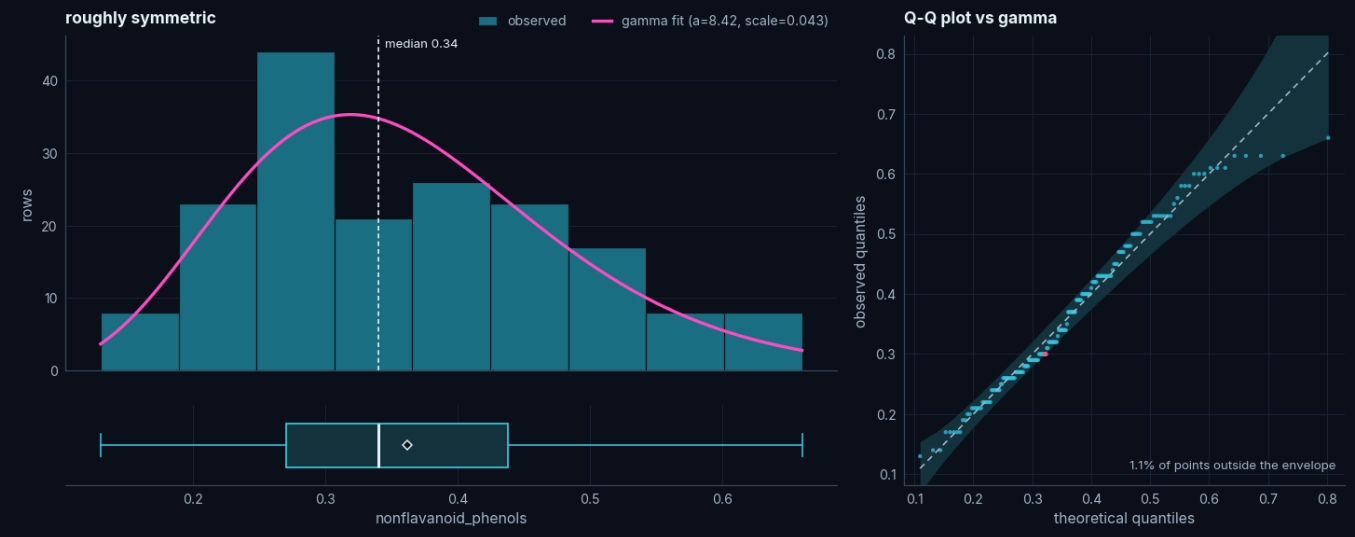
#	DISTRIBUTION	PARAMETERS	AIC	ΔAIC	WEIGHT	BIC	KS D	KS P
1	gamma	a=8.42, scale=0.043	-246.51	0.0	0.33	-240.146	0.081 (poor)	0.19
2	chi-squared	df=16.8, scale=0.0215	-246.51	0.0	0.33	-240.146	0.081 (poor)	0.19
3	beta	a=5.32, b=9.35, scale=1	-245.507	1.0	0.20	-239.144	0.097 (poor)	0.066

nonflavanoid_phenols · distribution

FIGURE // 11_nonflavanoid_phenols_dist.png

nonflavanoid_phenols

numeric_continuous · n = 178 · mean 0.3619 · median 0.34 · sd 0.1245 · skew 0.45 · not normal



Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

`proanthocyanins` **· numeric_continuous**

float64 · float64 dtype: 101 distinct real values

Decision path: any non-missing value? → a single distinct value? → how is it stored? (float64) → only 0 and 1? → dates written as numbers (YYYYMMDD, Unix time…)? → an identifier by its name? → unique, consecutive or sorted integers? → integers that label categories (codes)? → a small scale of consecutive integers? → integers? → a handful of distinct real values? → **real numbers yes: numeric_continuous** (certain: decided on every value)

PROFILE	
type	numeric_continuous
dtype	float64
values	178
missing	0 (0.0%)
distinct	101 (56.7%)
memory	1.4 KB
mode	1.35 ×9

CENTER	
mean	1.591
95% CI of mean	[1.506, 1.676]
median	1.555
95% CI of median	[1.44, 1.65]
trimmed mean	1.564
geometric mean	1.483
harmonic mean	1.362

SPREAD	
std	0.5724
variance	0.3276
sem	0.0429
IQR	0.7
MAD	0.38
CV	0.3598
range	3.17

QUANTILES	
min	0.41
p1	0.42
p5	0.73
p25	1.25
p50	1.555
p75	1.95
p95	2.709
p99	3.034
max	3.58
sum	283.2

SHAPE	
skewness	0.517
excess kurtosis	0.555
KDE modes	1
bimodality coef.	0.351
decimals	2
heaping	×1.2

OUTLIERS & COUNTS	
IQR fences	[0.2, 3]
outside fences	2 (1.1%)
|z| > 3	1 (0.6%)
modified z > 3.5	1 (0.6%)
zeros	0
negatives	0
infinite	0

ROW ORDER	
monotonic	no
lag-1 autocorrelation	0.385 (p 3e-07)

Normality: not normal (n = 178)

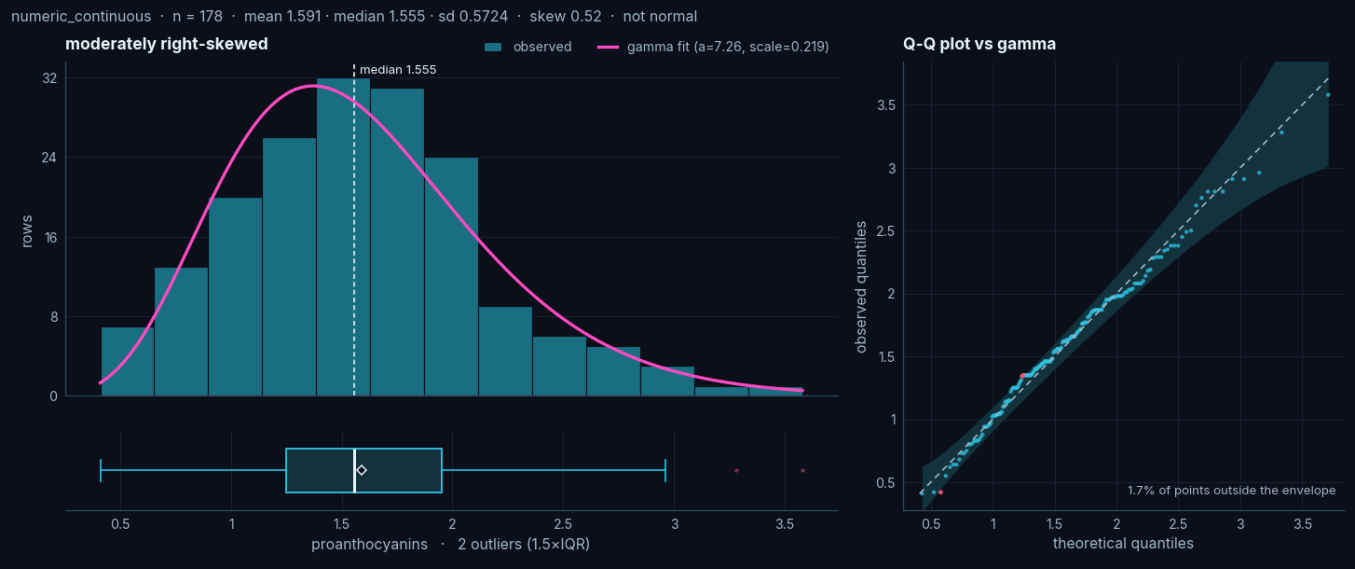
TEST	STATISTIC	P-VALUE	AT ALPHA 0.05
Shapiro-Wilk	0.9807	0.014	rejects normality
D'Agostino K^2	9.656	0.008	rejects normality
Jarque-Bera	9.697	0.0078	rejects normality
Anderson-Darling	0.6998	0.07	consistent with normal

Best fits (continuous, 13 candidates fitted on 178 values)

#	DISTRIBUTION	PARAMETERS	AIC	ΔAIC	WEIGHT	BIC	KS D	KS P
1	gamma	a=7.26, scale=0.219	304.626	0.0	0.37	310.99	0.077 (fair)	0.23
2	chi-squared	df=14.5, scale=0.11	304.626	0.0	0.37	310.99	0.077 (fair)	0.23
3	Weibull	c=2.97, scale=1.78	306.574	1.9	0.14	312.938	0.059 (fair)	0.55

proanthocyanins · distribution

FIGURE // 12_proanthocyanins_dist.png



Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

`color_intensity` · **numeric_continuous**

float64 · float64 dtype: 132 distinct real values

Decision path: any non-missing value? → a single distinct value? → how is it stored? (float64) → only 0 and 1? → dates written as numbers (YYYYMMDD, Unix time…)? → an identifier by its name? → unique, consecutive or sorted integers? → integers that label categories (codes)? → a small scale of consecutive integers? → integers? → a handful of distinct real values? → **real numbers yes: numeric_continuous** (certain: decided on every value)

PROFILE	
type	numeric_continuous
dtype	float64
values	178
missing	0 (0.0%)
distinct	132 (74.2%)
memory	1.4 KB
mode	3.8 ×4

CENTER	
mean	5.058
95% CI of mean	[4.715, 5.401]
median	4.69
95% CI of median	[4.28, 5.1]
trimmed mean	4.828
geometric mean	4.563
harmonic mean	4.106

SPREAD	
std	2.318
variance	5.374
sem	0.1738
IQR	2.98
MAD	1.51
CV	0.4583
range	11.72

QUANTILES	
min	1.28
p1	1.863
p5	2.114
p25	3.22
p50	4.69
p75	6.2
p95	9.598
p99	11.02
max	13
sum	900.3

SHAPE	
skewness	0.869
excess kurtosis	0.382
KDE modes	1
bimodality coef.	0.511
decimals	2

OUTLIERS & COUNTS	
IQR fences	[-1.25, 10.67]
outside fences	4 (2.2%)
|z| > 3	1 (0.6%)
modified z > 3.5	1 (0.6%)
zeros	0
negatives	0
infinite	0

ROW ORDER	
monotonic	no
lag-1 autocorrelation	0.745 (p 3e-23)

Normality: not normal (n = 178)

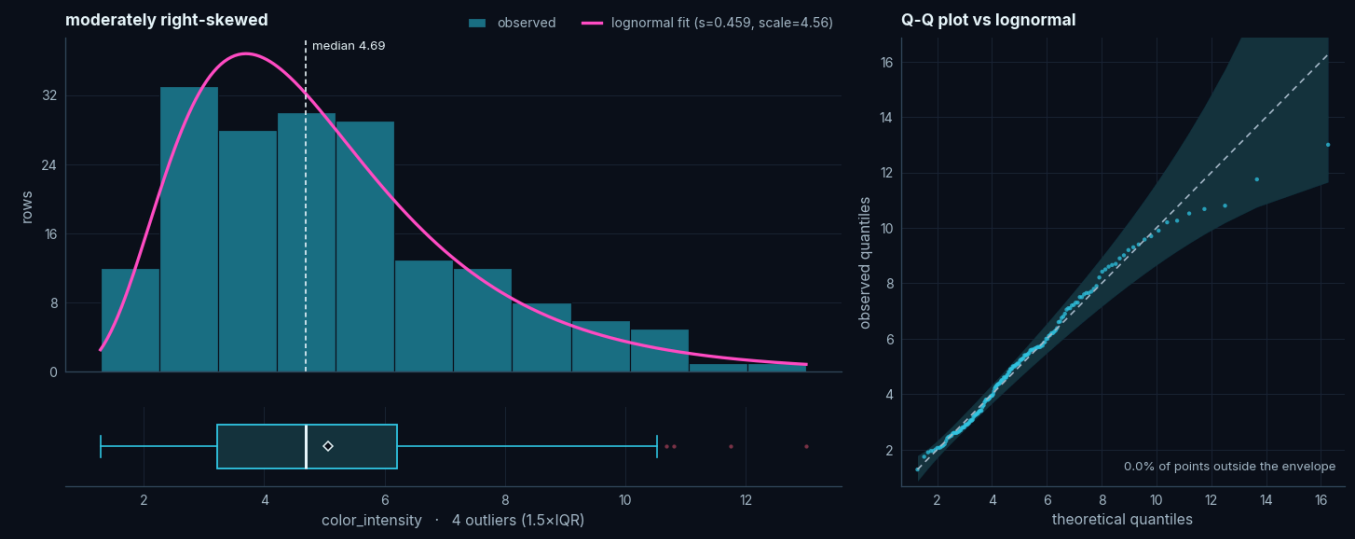
TEST	STATISTIC	P-VALUE	AT ALPHA 0.05
Shapiro-Wilk	0.9403	9e-07	rejects normality
D'Agostino K^2	19.68	5e-05	rejects normality
Jarque-Bera	22.85	1e-05	rejects normality
Anderson-Darling	2.844	0.01	rejects normality

Best fits (continuous, 13 candidates fitted on 178 values)

#	DISTRIBUTION	PARAMETERS	AIC	ΔAIC	WEIGHT	BIC	KS D	KS P
1	lognormal	s=0.459, scale=4.56	772.26	0.0	0.58	778.624	0.048 (fair)	0.78
2	gamma	a=5.02, scale=1.01	774.27	2.0	0.21	780.634	0.061 (fair)	0.5
3	chi-squared	df=10, scale=0.504	774.27	2.0	0.21	780.634	0.061 (fair)	0.5

color_intensity · distribution

FIGURE // 13_color_intensity_dist.png



Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

hue · **numeric_continuous**

float64 · float64 dtype: 78 distinct real values

Decision path: any non-missing value? → a single distinct value? → how is it stored? (float64) → only 0 and 1? → dates written as numbers (YYYYMMDD, Unix time…)? → an identifier by its name? → unique, consecutive or sorted integers? → integers that label categories (codes)? → a small scale of consecutive integers? → integers? → a handful of distinct real values? → **real numbers yes: numeric_continuous** (certain: decided on every value)

PROFILE	
type	numeric_continuous
dtype	float64
values	178
missing	0 (0.0%)
distinct	78 (43.8%)
memory	1.4 KB
mode	1.04 ×8

CENTER	
mean	0.9574
95% CI of mean	[0.9236, 0.9913]
median	0.965
95% CI of median	[0.92, 1.03]
trimmed mean	0.9586
geometric mean	0.9288
harmonic mean	0.8986

SPREAD	
std	0.2286
variance	0.05224
sem	0.01713
IQR	0.3375
MAD	0.165
CV	0.2387
range	1.23

QUANTILES	
min	0.48
p1	0.5477
p5	0.57
p25	0.7825
p50	0.965
p75	1.12
p95	1.285
p99	1.427
max	1.71
sum	170.4

SHAPE	
skewness	0.0211
excess kurtosis	-0.344
KDE modes	1
bimodality coef.	0.37
decimals	2
heaping	×1.0

OUTLIERS & COUNTS	
IQR fences	[0.2762, 1.626]
outside fences	1 (0.6%)
|z| > 3	1 (0.6%)
modified z > 3.5	0 (0.0%)
zeros	0
negatives	0
infinite	0

ROW ORDER	
monotonic	no
lag-1 autocorrelation	0.690 (p 3e-20)

Normality: not normal (n = 178)

TEST	STATISTIC	P-VALUE	AT ALPHA 0.05
Shapiro-Wilk	0.9813	0.017	rejects normality
D'Agostino K^2	1.039	0.59	consistent with normal
Jarque-Bera	1.018	0.6	consistent with normal
Anderson-Darling	0.8382	0.031	rejects normality

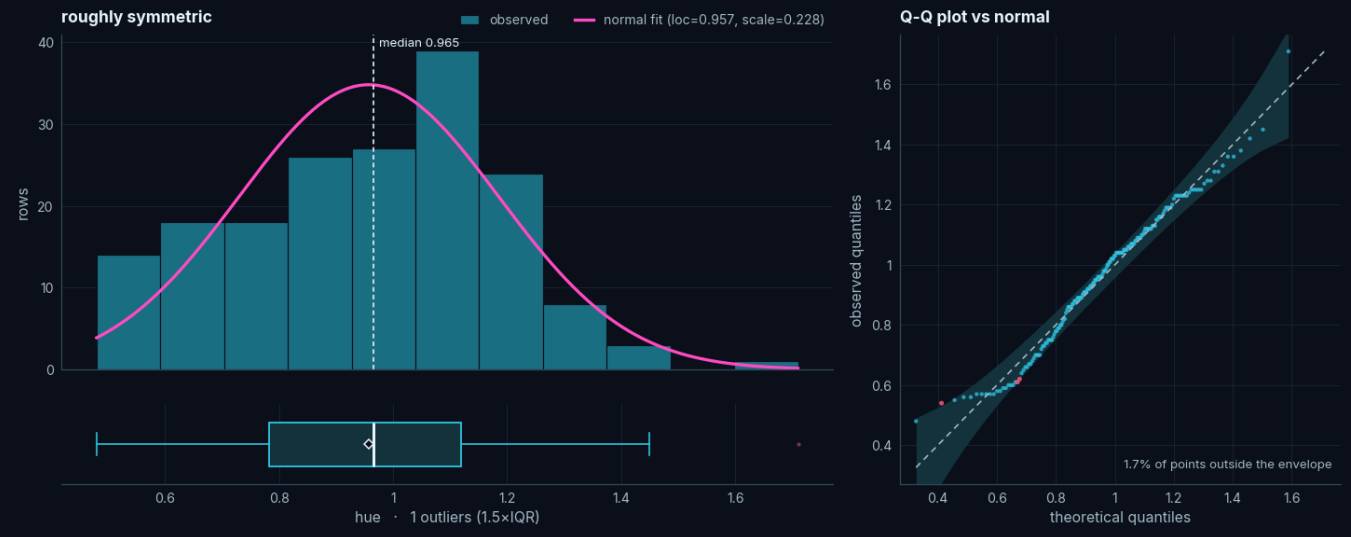
Best fits (continuous, 13 candidates fitted on 178 values)

#	DISTRIBUTION	PARAMETERS	AIC	ΔAIC	WEIGHT	BIC	KS D	KS P
1	normal	loc=0.957, scale=0.228	-17.2832	0.0	0.42	-10.9196	0.063 (fair)	0.47
2	Weibull	c=4.66, scale=1.05	-16.447	0.8	0.28	-10.0834	0.043 (fair)	0.88
3	Student t	df=4.67e+10, loc=0.957, scale=0.228	-15.2832	2.0	0.15	-5.73786	0.063 (fair)	0.47

hue · distribution FIGURE // 14_hue_dist.png

hue

numeric_continuous · n = 178 · mean 0.9574 · median 0.965 · sd 0.2286 · skew 0.02 · not normal



Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

`od280/od315_of_diluted_wines` **· numeric_continuous** —

float64 · float64 dtype: 122 distinct real values

Decision path: any non-missing value? → a single distinct value? → how is it stored? (float64) → only 0 and 1? → dates written as numbers (YYYYMMDD, Unix time…)? → an identifier by its name? → unique, consecutive or sorted integers? → integers that label categories (codes)? → a small scale of consecutive integers? → integers? → a handful of distinct real values? → **real numbers yes: numeric_continuous** (certain: decided on every value)

PROFILE	
type	numeric_continuous
dtype	float64
values	178
missing	0 (0.0%)
distinct	122 (68.5%)
memory	1.4 KB
mode	2.87 ×5

CENTER	
mean	2.612
95% CI of mean	[2.507, 2.717]
median	2.78
95% CI of median	[2.63, 2.87]
trimmed mean	2.631
geometric mean	2.504
harmonic mean	2.387

SPREAD	
std	0.71
variance	0.5041
sem	0.05322
IQR	1.232
MAD	0.52
CV	0.2719
range	2.73

QUANTILES	
min	1.27
p1	1.29
p5	1.462
p25	1.938
p50	2.78
p75	3.17
p95	3.58
p99	3.843
max	4
sum	464.9

SHAPE	
skewness	-0.307
excess kurtosis	-1.09
KDE modes	2
bimodality coef.	0.557
decimals	2
heaping	×1.0

OUTLIERS & COUNTS	
IQR fences	[0.08875, 5.019]
outside fences	0 (0.0%)
|z| > 3	0 (0.0%)
modified z > 3.5	0 (0.0%)
zeros	0
negatives	0
infinite	0

ROW ORDER	
monotonic	no
lag-1 autocorrelation	0.763 (p 2e-24)

Normality: not normal (n = 178)

TEST	STATISTIC	P-VALUE	AT ALPHA 0.05
Shapiro-Wilk	0.945	2e-06	rejects normality
D'Agostino K^2	47.49	5e-11	rejects normality
Jarque-Bera	11.56	0.0031	rejects normality
Anderson-Darling	3.496	0.01	rejects normality

Best fits (continuous, 13 candidates fitted on 178 values)

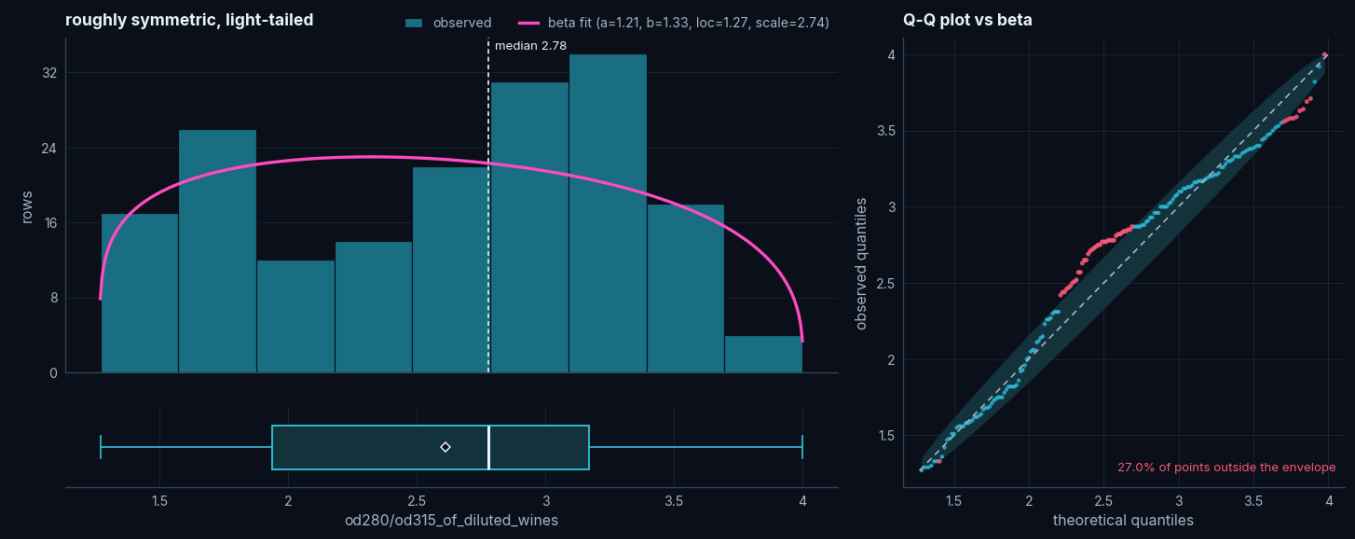
#	DISTRIBUTION	PARAMETERS	AIC	ΔAIC	WEIGHT	BIC	KS D	KS P
1	beta	a=1.21, b=1.33, loc=1.27, scale=2.74	358.748	0.0	0.80	371.475	0.129 (poor)	0.0047
2	uniform	loc=1.27, scale=2.73	361.531	2.8	0.20	367.895	0.111 (poor)	0.023
3	Weibull	c=4.38, scale=2.88	377.691	18.9	0.00	384.055	0.107 (poor)	0.031

od280/od315_of_diluted_wines · distribution

FIGURE // 15_od280_od315_of_diluted_wines_dist.png

od280/od315_of_diluted_wines

numeric_continuous · n = 178 · mean 2.612 · median 2.78 · sd 0.71 · skew -0.31 · not normal



Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

proline · numeric_continuous · 1 alert

float64 · float64 dtype: integer-valued with many levels (121)

Decision path: any non-missing value? → a single distinct value? → how is it stored? (float64) → only 0 and 1? → dates written as numbers (YYYYMMDD, Unix time…)? → an identifier by its name? → unique, consecutive or sorted integers? → integers that label categories (codes)? → a small scale of consecutive integers? → **integers? yes: numeric_continuous** (certain: decided on every value)

- INFO heaping: 83% of values end in 0 or 5 (expected ~20%) → rounded / self-reported values?

PROFILE	
type	numeric_continuous
dtype	float64
values	178
missing	0 (0.0%)
distinct	121 (68.0%)
memory	1.4 KB
mode	680.0 ×5

CENTER	
mean	746.9
95% CI of mean	[700.3, 793.5]
median	673.5
95% CI of median	[625, 718]
trimmed mean	719.3
geometric mean	685.6
harmonic mean	630.6

SPREAD	
std	314.9
variance	99,166.7
sem	23.6
IQR	484.5
MAD	202.5
CV	0.4216
range	1,402

QUANTILES	
min	278
p1	306.9
p5	354.6
p25	500.5
p50	673.5
p75	985
p95	1,297.2
p99	1,522.4
max	1,680
sum	132,947

SHAPE	
skewness	0.768
excess kurtosis	-0.248
KDE modes	1
bimodality coef.	0.567
decimals	0
heaping	×4.2

OUTLIERS & COUNTS	
IQR fences	[-226.2, 1,711.8]
outside fences	0 (0.0%)
|z| > 3	0 (0.0%)
modified z > 3.5	0 (0.0%)
zeros	0
negatives	0
infinite	0

ROW ORDER	
monotonic	no
lag-1 autocorrelation	0.736 (p 9e-23)

Normality: not normal (n = 178)

TEST	STATISTIC	P-VALUE	AT ALPHA 0.05
Shapiro-Wilk	0.9312	2e-07	rejects normality
D'Agostino K^2	15.52	4e-04	rejects normality
Jarque-Bera	17.76	1e-04	rejects normality
Anderson-Darling	4.1	0.01	rejects normality

Best fits (continuous, 13 candidates fitted on 178 values)

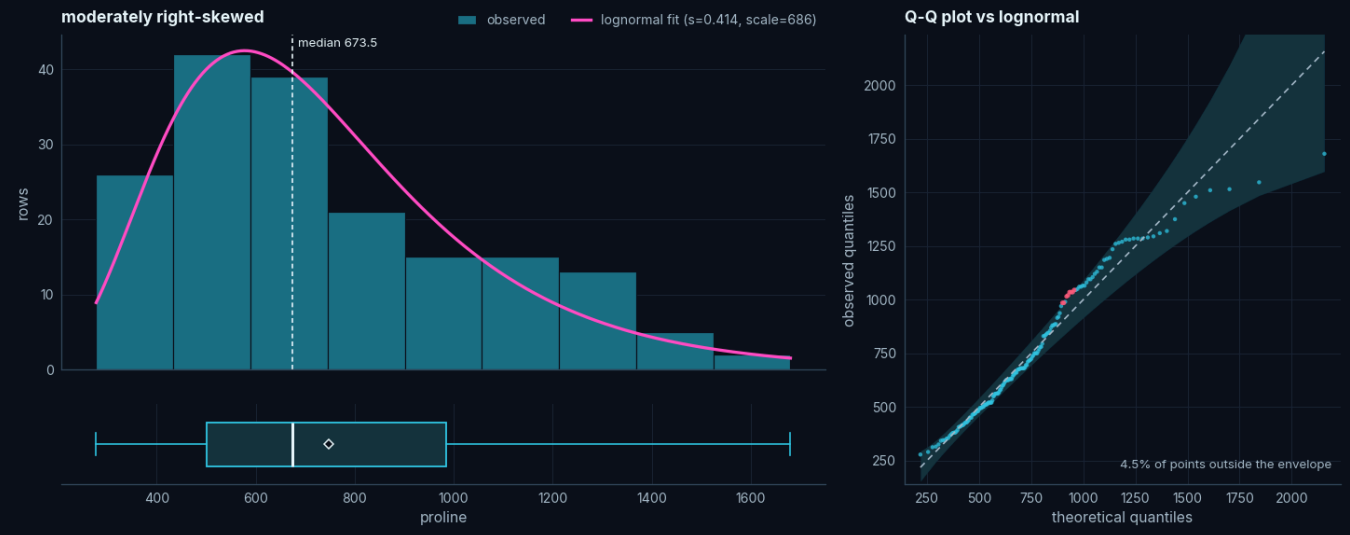
#	DISTRIBUTION	PARAMETERS	AIC	ΔAIC	WEIGHT	BIC	KS D	KS P
1	lognormal	s=0.414, scale=686	2,519.9	0.0	0.68	2,526.3	0.070 (fair)	0.32
2	beta	a=1.17, b=2.2, loc=277, scale=1.4e+03	2,522.4	2.5	0.20	2,535.1	0.092 (poor)	0.094
3	gamma	a=6, scale=124	2,524.8	4.9	0.06	2,531.2	0.080 (poor)	0.19

proline · distribution

FIGURE // 16_proline_dist.png

proline

numeric_continuous · n = 178 · mean 746.9 · median 673.5 · sd 314.9 · skew 0.77 · not normal



Histogram in rows; the curve is the count expected under the best fit (MLE, lowest AIC). Q-Q envelope: 95% pointwise band for a sample of this size.

`cultivar` **· categorical** —

category · category dtype (3 levels)

Decision path: any non-missing value? → a single distinct value? → **how is it stored? (category) yes: categorical** (certain: decided on every value)

PROFILE	
type	categorical
dtype	category
values	178
missing	0 (0.0%)
distinct	3 (1.7%)
memory	526 B
mode	class_1 ×71

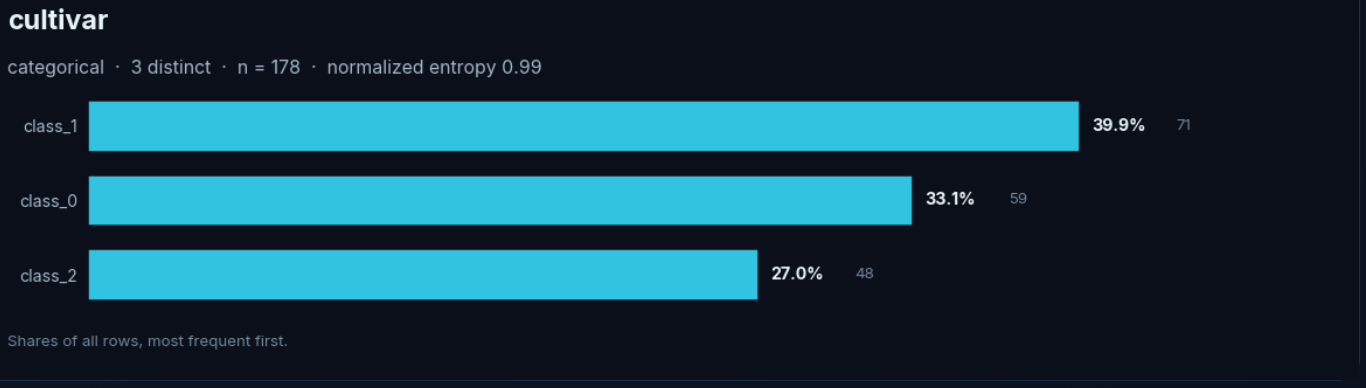
LABELS	
cardinality	3
entropy	1.57 bits (0.99 of max)
Gini impurity	0.658
imbalance	×1.479
rare labels	0 (0.0% of rows)
uniform? chi²	p 0.11

Most frequent values

VALUE	COUNT	SHARE
`class_1`	71	39.9%
`class_0`	59	33.1%
`class_2`	48	27.0%

cultivar · frequencies

FIGURE // 17_cultivar_bar.png



11 / RUN

Run.

Source	pandas.DataFrame · read in 0.0s
Processes	1 for the analysis, 1 for the charts (5 cores available)
Time	14.2s (load 0.0s, columns 1.4s, associations 0.3s, quality 0.0s, advanced 1.2s, charts 11.3s)
Significance	alpha 0.05, Benjamini-Hochberg false discovery rate per family of tests
Versions	elessar 1.1.0 · Python 3.13.5 · pandas 2.2.3 · numpy 2.3.5 · scipy 1.17.0